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FOR WOMEN**

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**Academic year : 2019-2020**

**LABORATORY MANUAL OF**

**19A04303: BASIC SIMULATION LAB**

For

**II.B.Tech - I Sem**

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## **GENERATION OF VARIOUS SIGNALS & SEQUENCES**

**AIM:** To generate various signals and sequences: Periodic and Aperiodic, Unit impulse, Unit step, Square, Saw tooth, Triangular, Sinusoidal, Ramp, Sinc function.

**APPARATUS:** Computer

MATLAB 7.8.0.347 (R2009a) Software.

### **PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

### **PROGRAM:**

```
t=-10:0.1:10

%-----Unit Impulse-----%

x1=1;

x2=0;

x =x1.*(t==0)    %generate Unit Impulse Signal

figure(1), plot(t,x); % plot the generated Unit Impulse Signal

xlabel('t'); ylabel('x(t)'); title('Unit impulse Signal');

%----- Unit Step signal-----%

x1 =1;

x2 = 0;
```

```
x = x1.*(t>=0)+x2.*(t<0);    %generate Unit Step Signal

figure(2); plot(t,x);        %Plot the generated unit step signal

xlabel('t'); ylabel('x(t)'); title('Unit Step Signal');

%----- Ramp Signal-----%

x1 = t;

x2 = 0;

x = x1.*(t>=0)+x2.*(t<0);    %generate unit ramp signal

figure(3);plot(t,x);        % plot the generated ramp signal

xlabel('t'); ylabel('x(t)'); title(' Ramp Signal');

%-----rectangular pulse-----%

t1=-10:0.01:10

a7=rectpuls(t1/10)

figure(4),plot(t1,a7), grid on,axis([-10 10 -2 2])

xlabel('time')

ylabel('magnitude')

title('rectangular pulse')

%----- triangular pulse-----%

t2=-10:0.1:10

a8=tripuls(t2)

figure(5),plot(t2,a8), grid on

title('triangular pulse')

xlabel('time')

ylabel('magnitude')

%----- Sinc Signal-----%

x = sinc(t);                  %generate sinc signal
```

```
figure(6);plot(t,x);    % plot the generated sinc signal

xlabel('t'); ylabel('x(t)'); title('sinc Signal');

%-----Sinusoidal Signal-----%

T = 2;                %declare time period

F = 1/T;              %compute frequency

x = sin(2*pi*F*t);    %generate sinusoidal signal

figure(7);plot(t,x);    % plot the generated sinusoidal signal

xlabel('t'); ylabel('x(t)'); title('sinusoidal Signal');

%----- Sawtooth signal-----%

fs = 10000;

t = 0:1/fs:1.5;

x = sawtooth(2*pi*50*t);    %generate sawtooth signal

figure(8);plot(t,x); axis([0 0.2 -1 1]); %plot the generated sawtooth signal

xlabel('t'); ylabel('x(t)'); title('sawtooth Signal');

%----- square signal-----%

t=(0:0.001:10);

sq_wave=square(4*pi*t);    %generate square signal

figure(9); plot(t,sq_wave); axis([0 5 -2 2])    %plot the generated square signal

xlabel('t'); ylabel('x(t)'); title('square signal');

%-----periodic triangle-----%

T=20;

x=linspace(0,5*T,10000);

y1=min(0.5,mod(x,T)/T);

y2=min(0.5,1-mod(x,T)/T);

figure(10),plot(x,y1+y2),grid on
```

```
xlabel('time')
ylabel('magnitude')
title('periodic triangle wave')

tc = gauspuls('cutoff',50e3,0.6,[],-40);

t = -tc : 1e-6 : tc;

yi = gauspuls(t,50e3,0.6);

figure(11),plot(t,yi),grid on

xlabel('time')
ylabel('magnitude')
title('Aperiodic triangle wave')

t=-10:0.1:10

%-----Unit Impulse-----%

x1=1;

x2=0;

x =x1.*(t==0)    %generate Unit Impulse Signal

figure(12), stem(t,x); % plot the generated Unit Impulse Signal

xlabel('t'); ylabel('x(t)'); title('Unit impulse Signal');

%----- Unit Step signal-----%

x1 =1;

x2 = 0;

x = x1.*(t>=0)+x2.*(t<0);    %generate Unit Step Signal

figure(13); stem(t,x);    %Plot the generated unit step signal

xlabel('t'); ylabel('x(t)'); title('Unit Step Signal');

%----- Ramp Signal-----%

x1 = t;
```

```
x2 = 0;

x = x1.*(t>=0)+x2.*(t<0);    %generate unit ramp signal

figure(14);stem(t,x);        % plot the generated ramp signal

xlabel('t'); ylabel('x(t)'); title(' Ramp Signal');

%-----rectangular pulse-----%

t1=-10:0.01:10

a7=rectpuls(t1/10)

figure(15),stem(t1,a7), grid on,axis([-10 10 -2 2])

xlabel('time')

ylabel('magnitude')

title('rectangular pulse')

%----- triangular pulse-----%

t2=-10:0.1:10

a8=tripuls(t2)

figure(16),stem(t2,a8), grid on

title('triangular pulse')

xlabel('time')

ylabel('magnitude')

%----- Sinc Signal-----%

x = sinc(t);                  %generate sinc signal

figure(17);stem(t,x);        % plot the generated sinc signal

xlabel('t'); ylabel('x(t)'); title('sinc Signal');

%-----Sinusoidal Signal-----%

T = 2;                        %declare time period

F = 1/T;                      %compute frequency
```

```
x = sin(2*pi*F*t);           %generate sinusoidal signal

figure(18);stem(t,x);        % plot the generated sinusoidal signal

xlabel('t'); ylabel('x(t)'); title('sinusoidal Signal');

%----- Sawtooth signal-----%

fs = 10000;

t = 0:1/fs:1.5;

x = sawtooth(2*pi*50*t);      %generate sawtooth signal

figure(19);stem(t,x); axis([0 0.2 -1 1]); %plot the generated sawtooth signal

xlabel('t'); ylabel('x(t)'); title('sawtooth Signal');

%----- square signal-----%

t=(0:0.001:10);

sq_wave=square(4*pi*t);       %generate square signal

figure(20); stem(t,sq_wave); axis([0 5 -2 2]) %plot the generated square signal

xlabel('t'); ylabel('x(t)'); title('square signal');

%-----periodic triangle-----%

T=20;

x=linspace(0,5*T,10000);

y1=min(0.5,mod(x,T)/T);

y2=min(0.5,1-mod(x,T)/T);

figure(21),stem(x,y1+y2),grid on

xlabel('time')

ylabel('magnitude')

title('periodic triangle wave')
```



```
tc = gauspuls('cutoff',50e3,0.6,[],-40);
```

```
t = -tc : 1e-6 : tc;
```

```
yi = gauspuls(t,50e3,0.6);
```

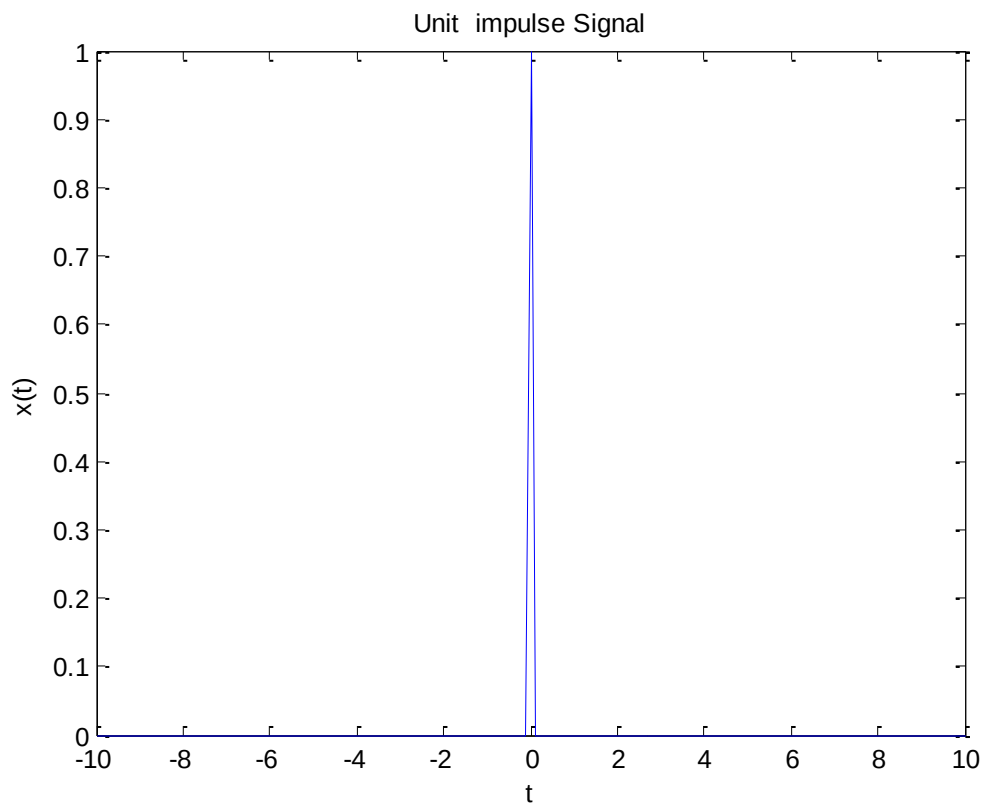
```
figure(22),stem(t,yi),grid on
```

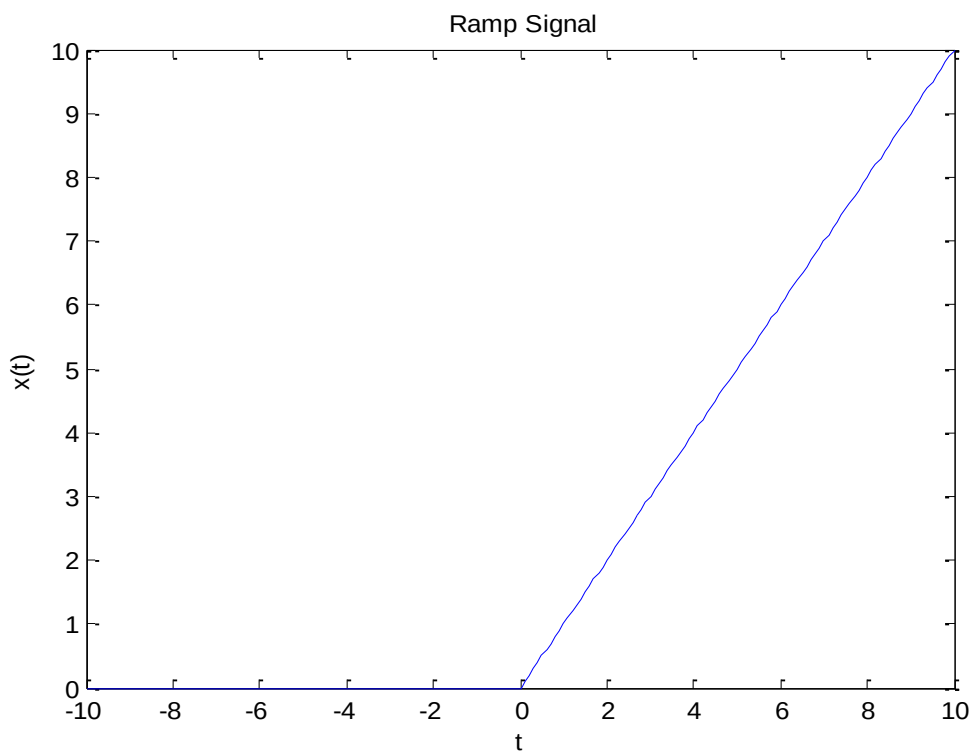
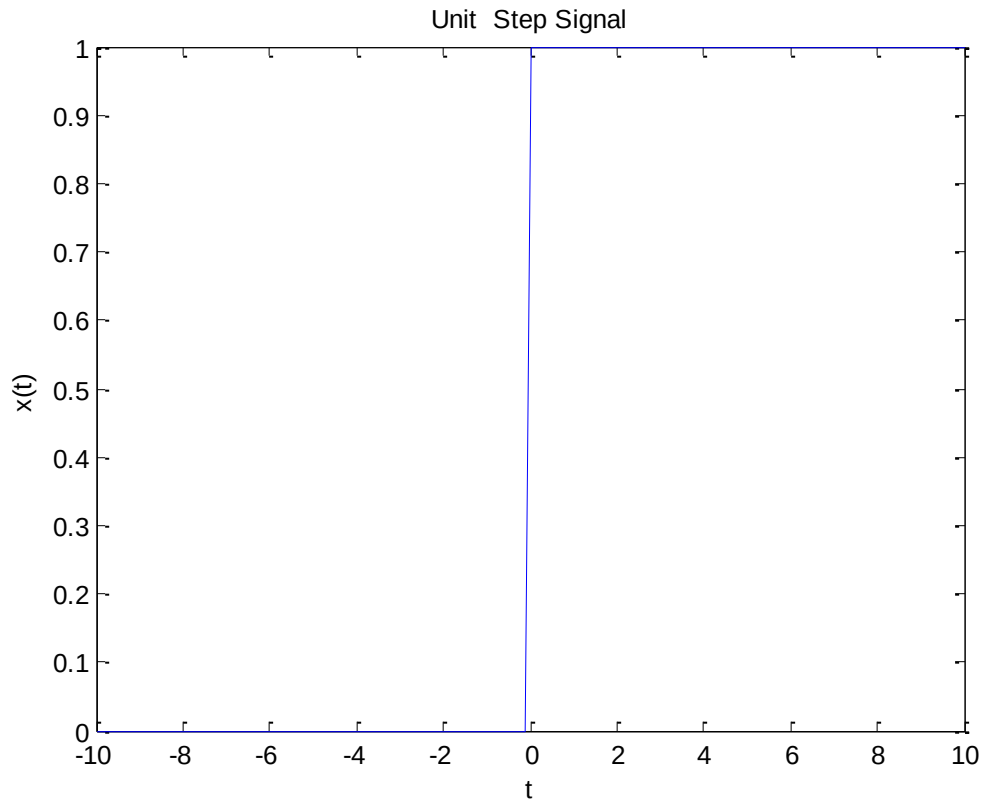
```
xlabel('time')
```

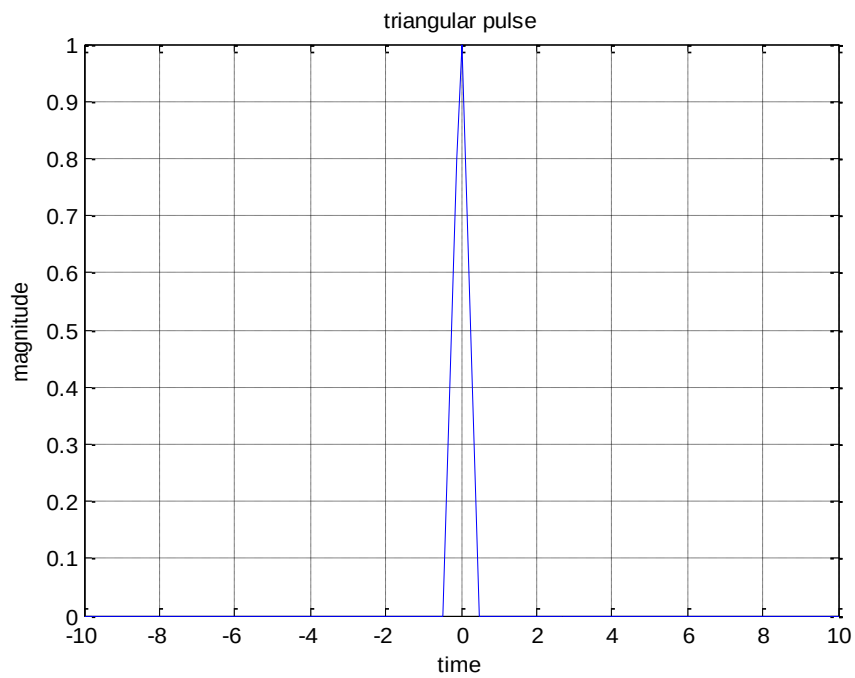
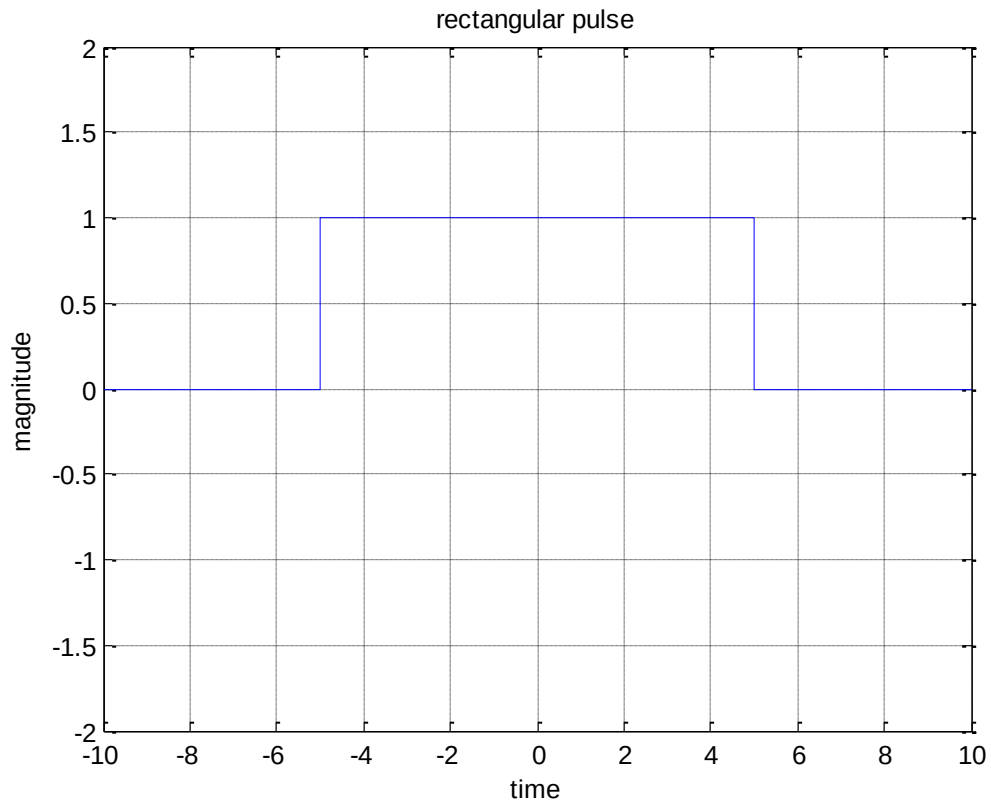
```
ylabel('magnitude')
```

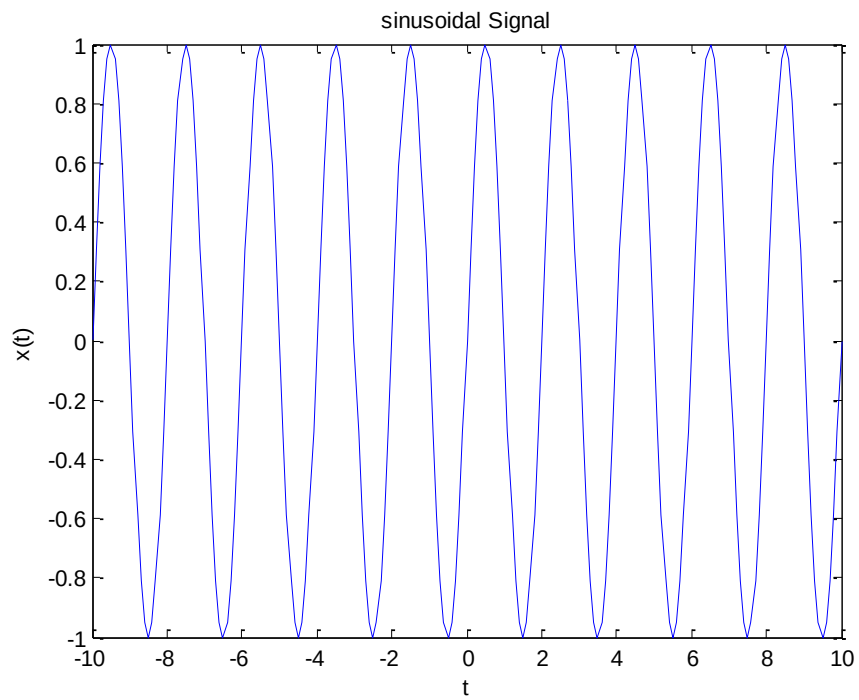
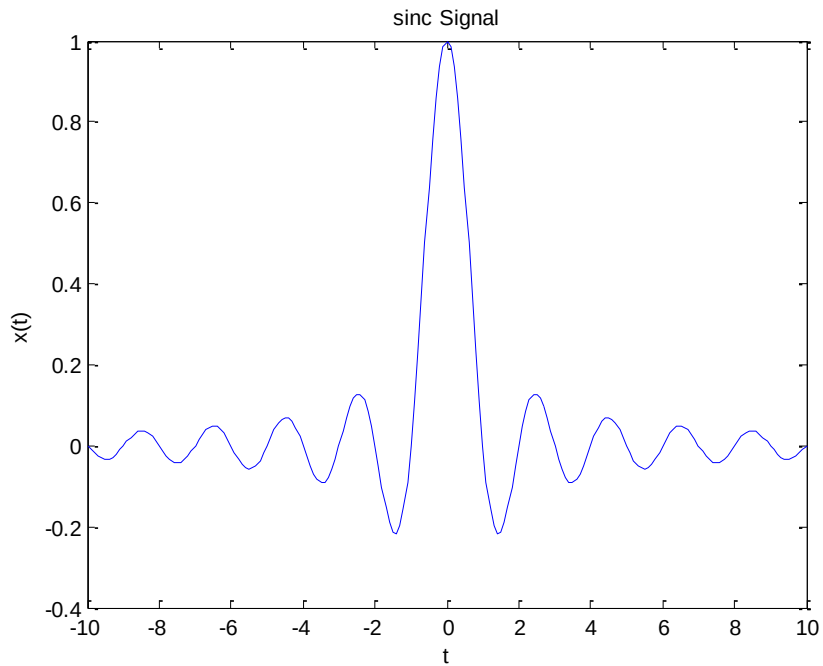
```
title('Aperiodic triangle wave')
```

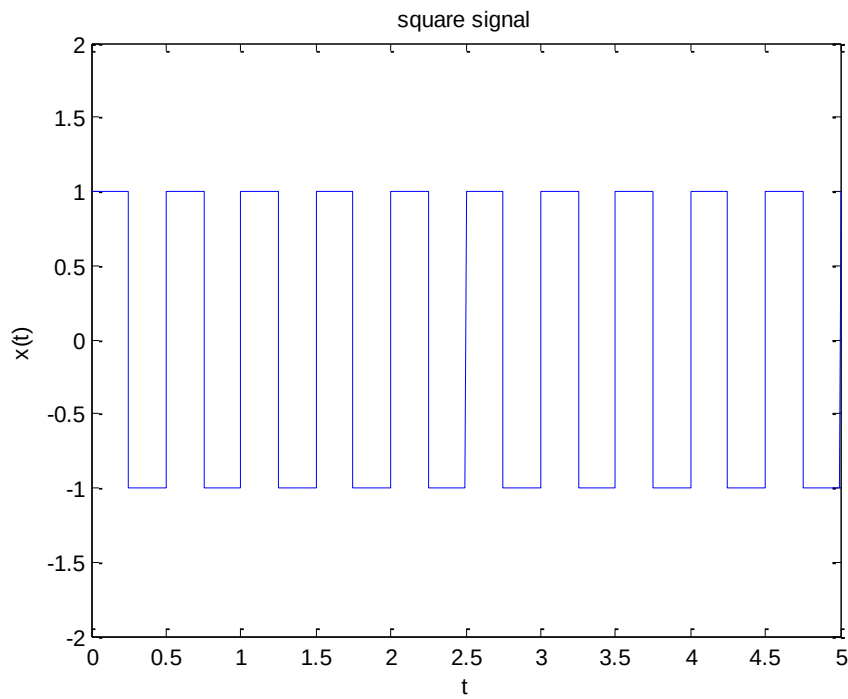
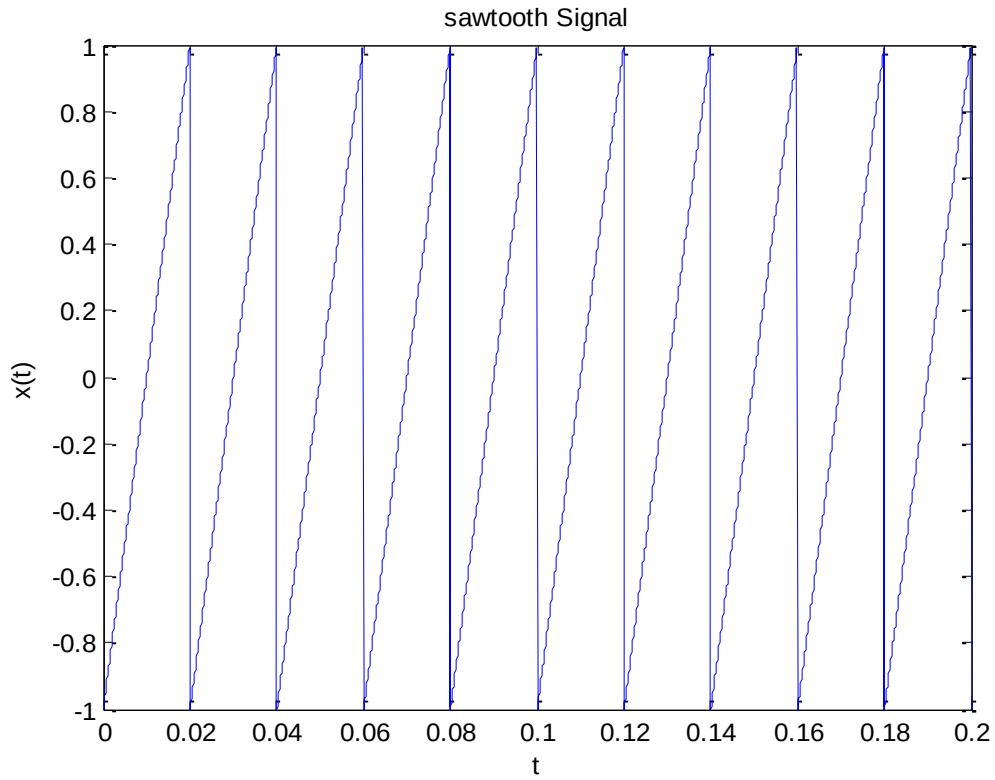
### **OUTPUT:**

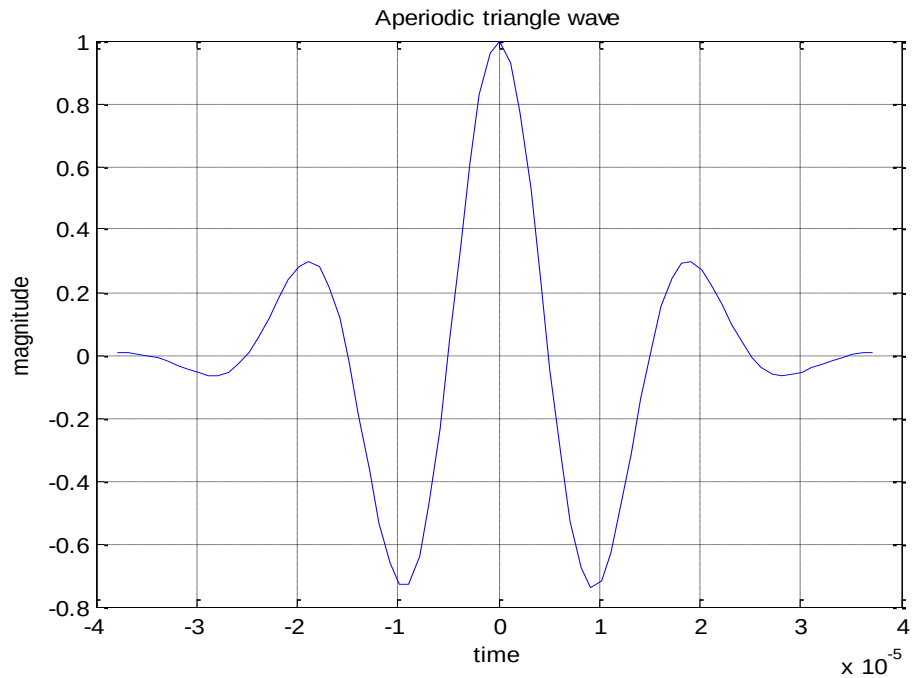
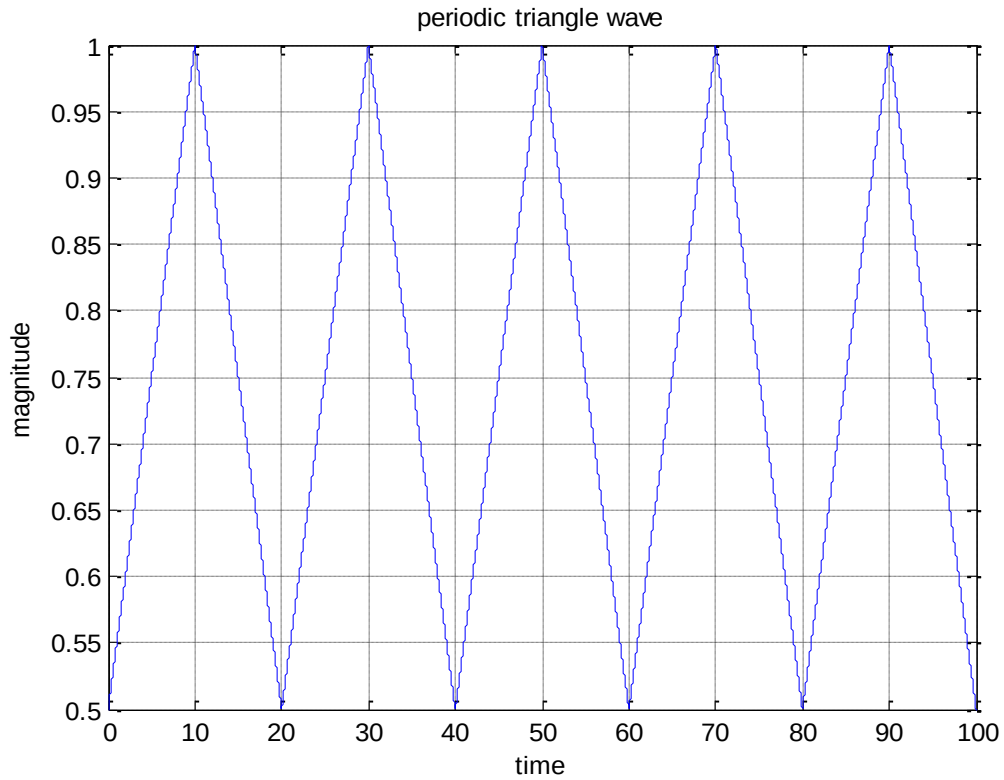


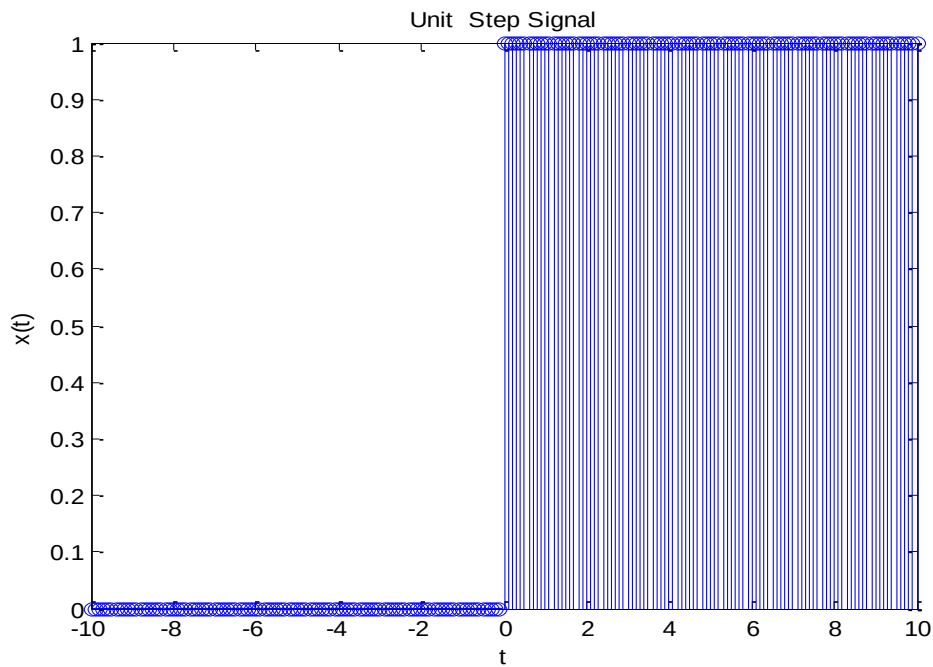
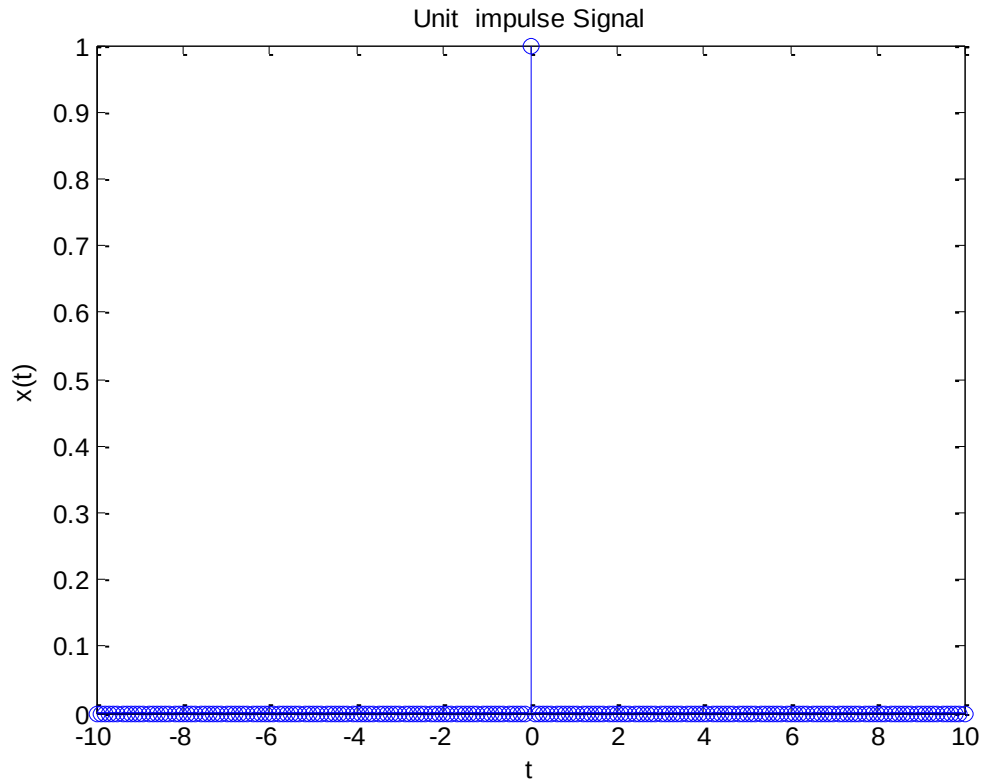


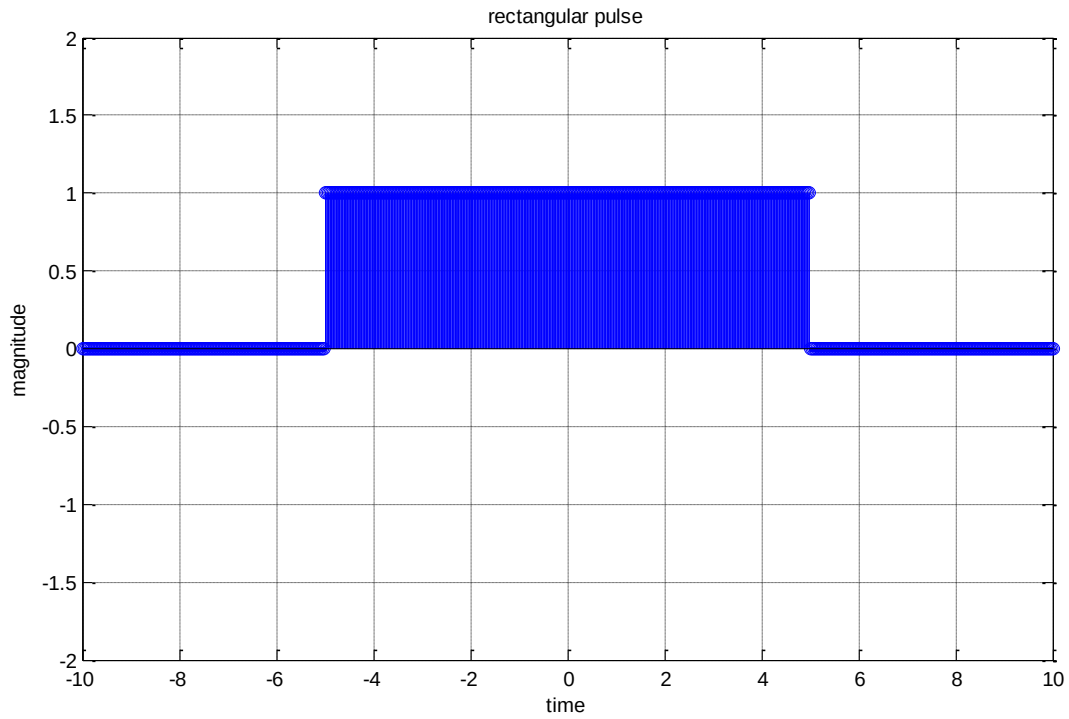
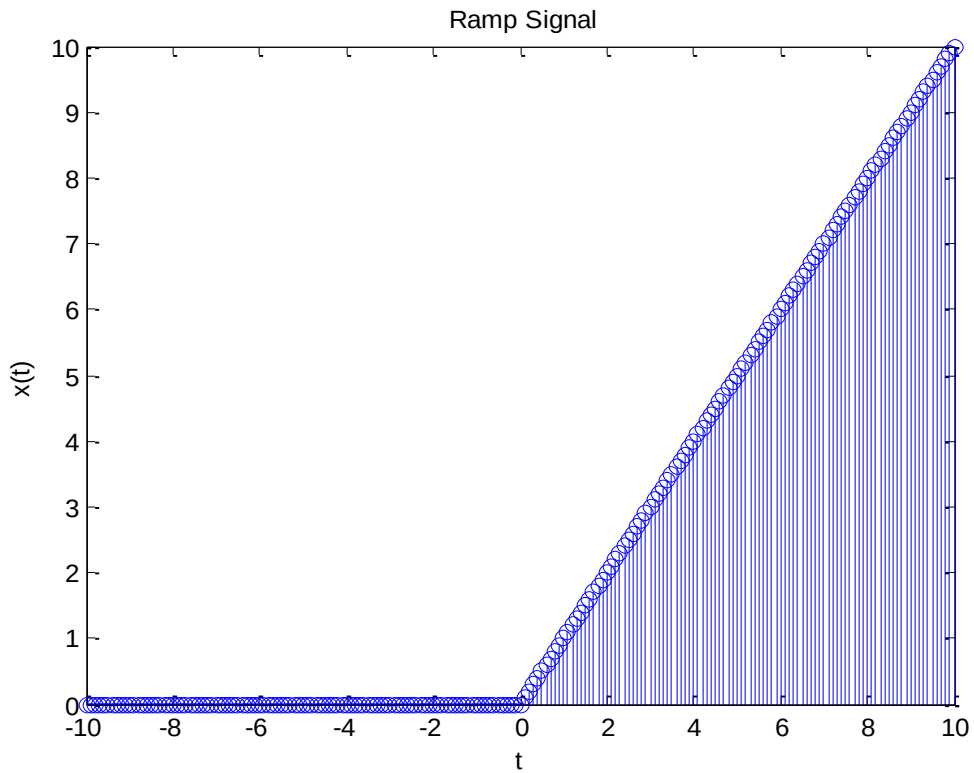




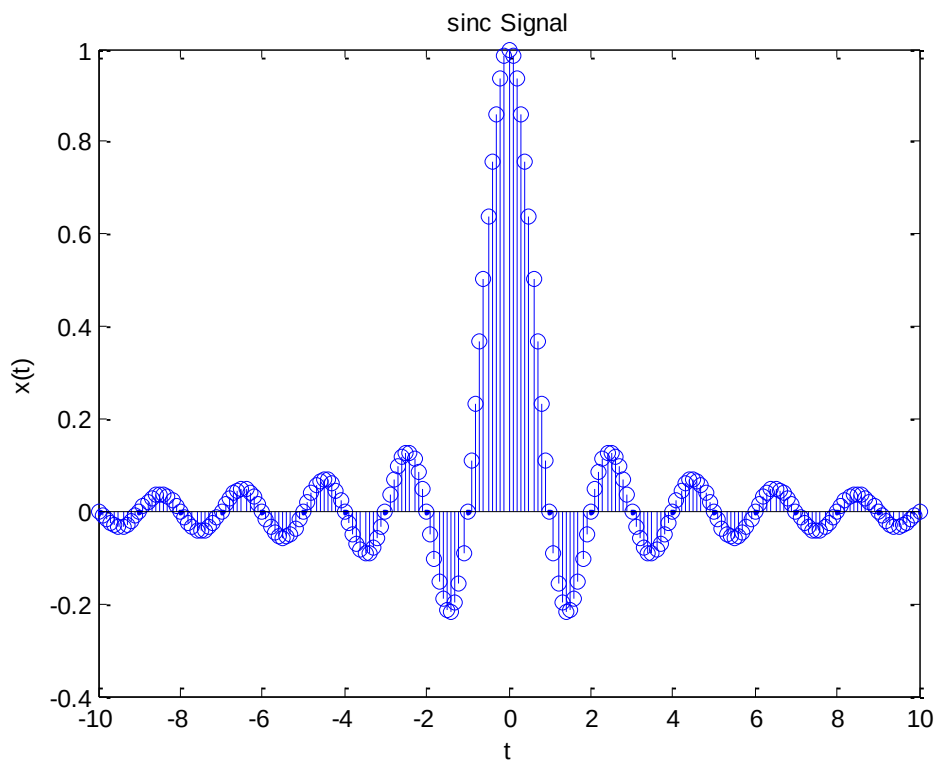
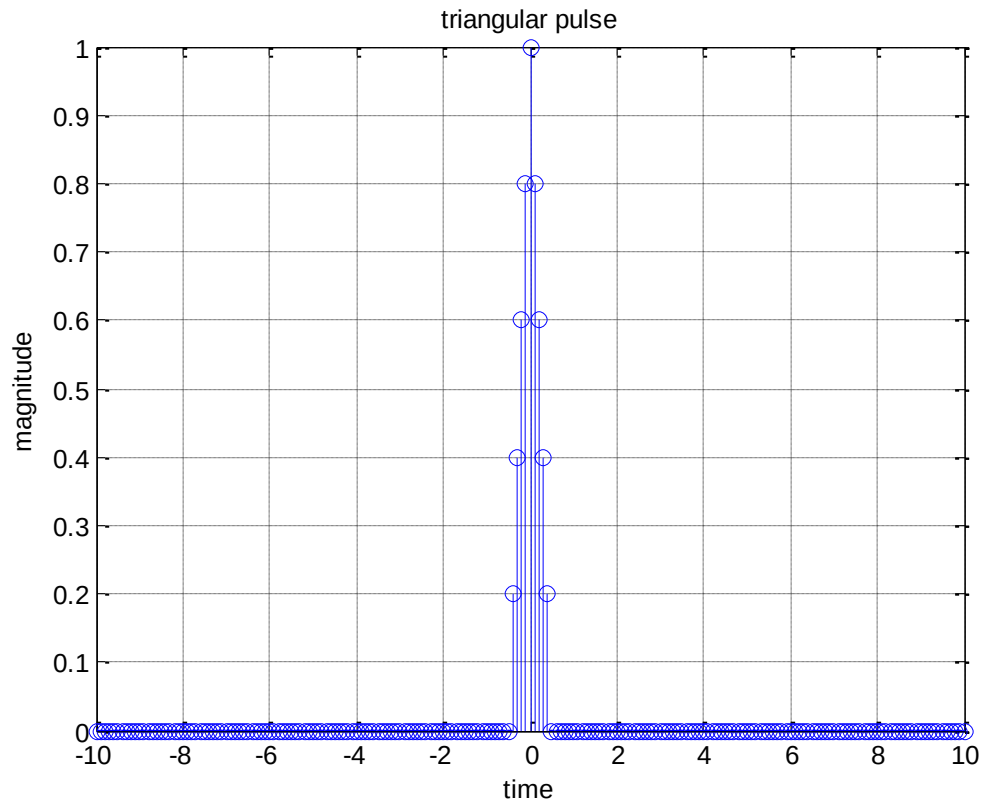


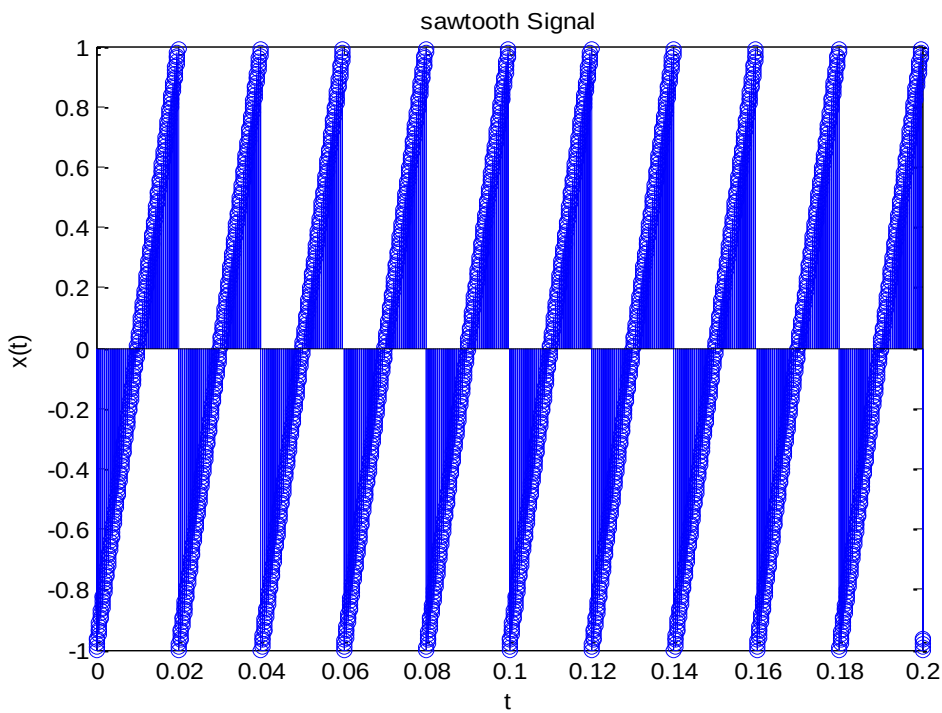
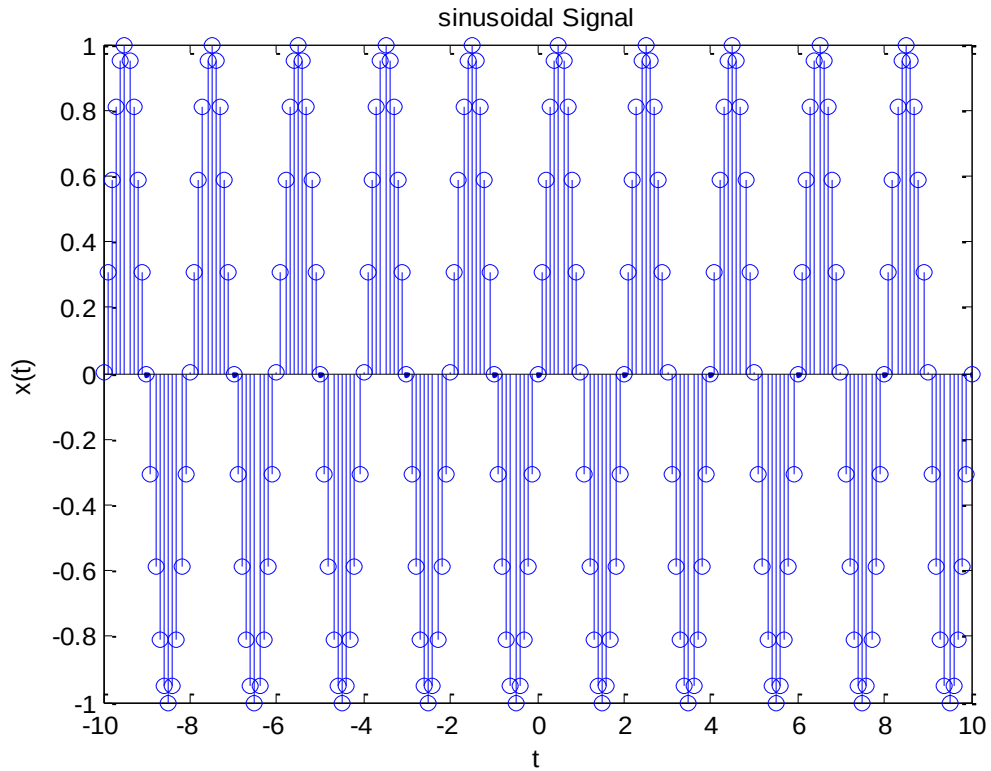


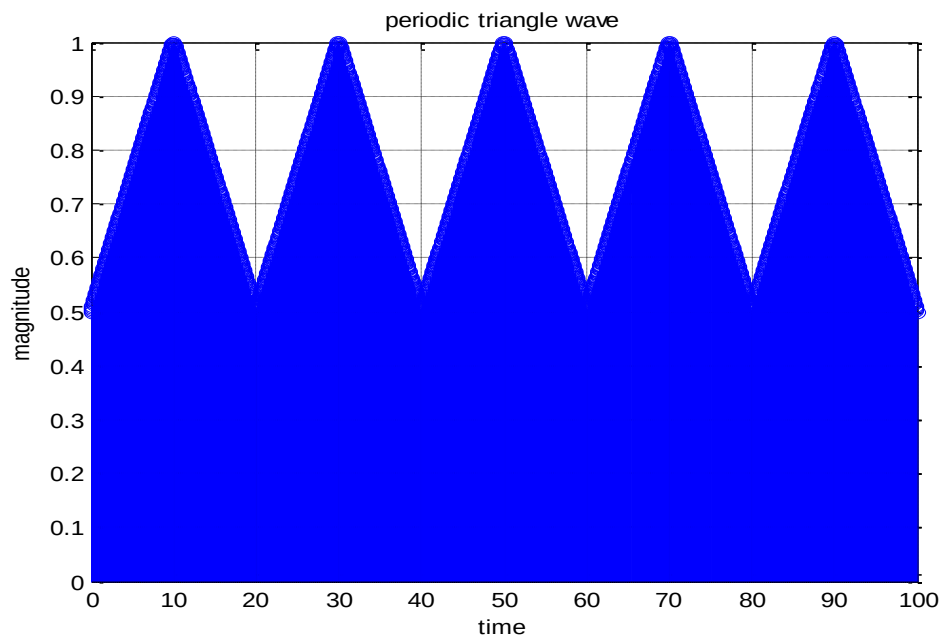
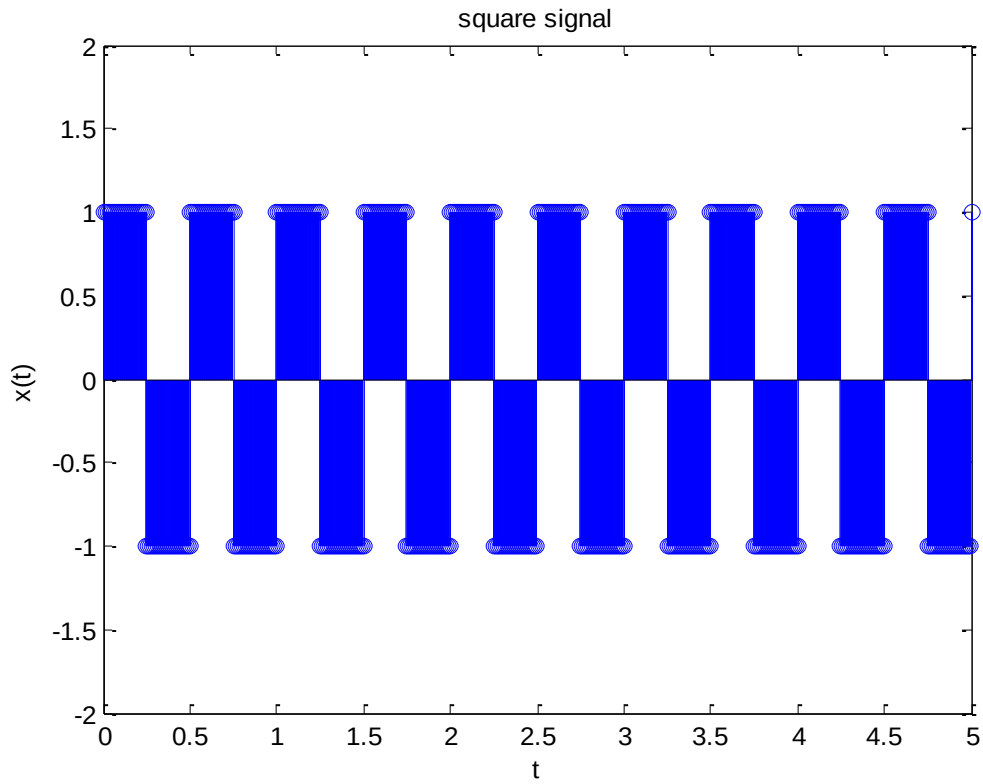


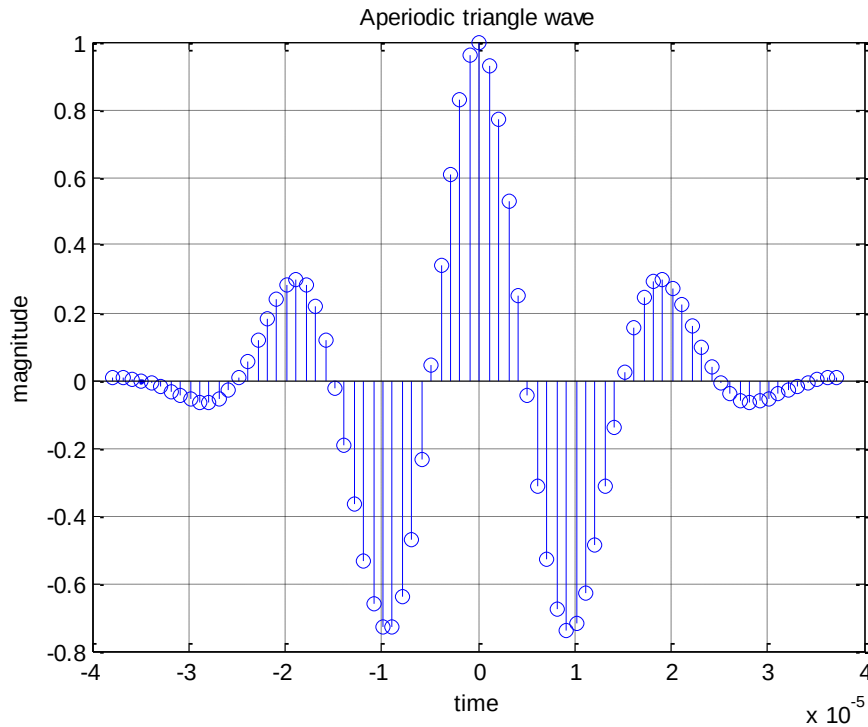












**RESULT:** Generation of various signals and sequences: Periodic And Aperiodic, Unit impulse, Unit step, Square, Saw tooth, Triangular, Sinusoidal, Ramp, Sinc function, is done using MATLAB

**PERFORM OPERATIONS ON VARIOUS SIGNALS & SEQUENCES,**  
**COMPUTATIONS OF ENERGY AND AVG POWER**

**AIM:** To perform operations on signals and sequences: addition ,multiplication ,scaling ,shifting , folding ,computation of energy and avg power

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

**PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

**PROGRAM:**

```
t=-10:0.1:10;
y1=sin(t);
y2=cos(t);
subplot(5,5,1),plot(t,y1),title('sinewave');grid on
subplot(5,5,2),plot(t,y2),title('coswave');grid on
%-----addition of two signals-----%
y3=y1+y2;
subplot(5,5,3),plot(t,y3),title('addition');grid on
%-----subtraction of two signals-----%
y4=y1-y2;
subplot(5,5,4),plot(t,y4),title('subtraction');grid on
%-----multiplication of two signals-----%
y5=y1.*y2;
subplot(5,5,5),plot(t,y5),title('multiplication');grid on
%-----time shifting-----%
y6=cos(t-5);
subplot(5,5,6),plot(t,y6),title('timeshifting(right)');grid on
y7=cos(t+5);
subplot(5,5,7),plot(t,y7),title('timeshifting(left)');grid on
%-----time reversal (folding) -----%
y8=sin(-t);
subplot(5,5,8),plot(t,y8),title('timefolding');grid on
```

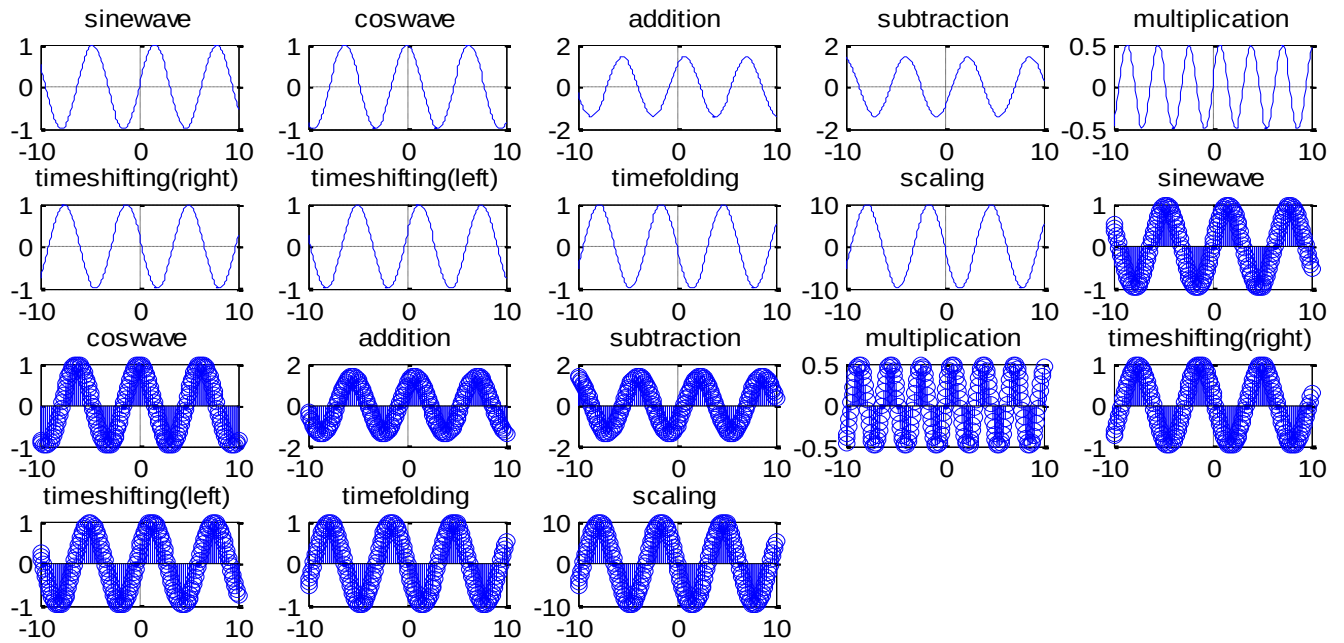
```
%-----scaling) -----%
y9=10*sin(-t);
subplot(5,5,9),plot(t,y9),title('scaling');grid on

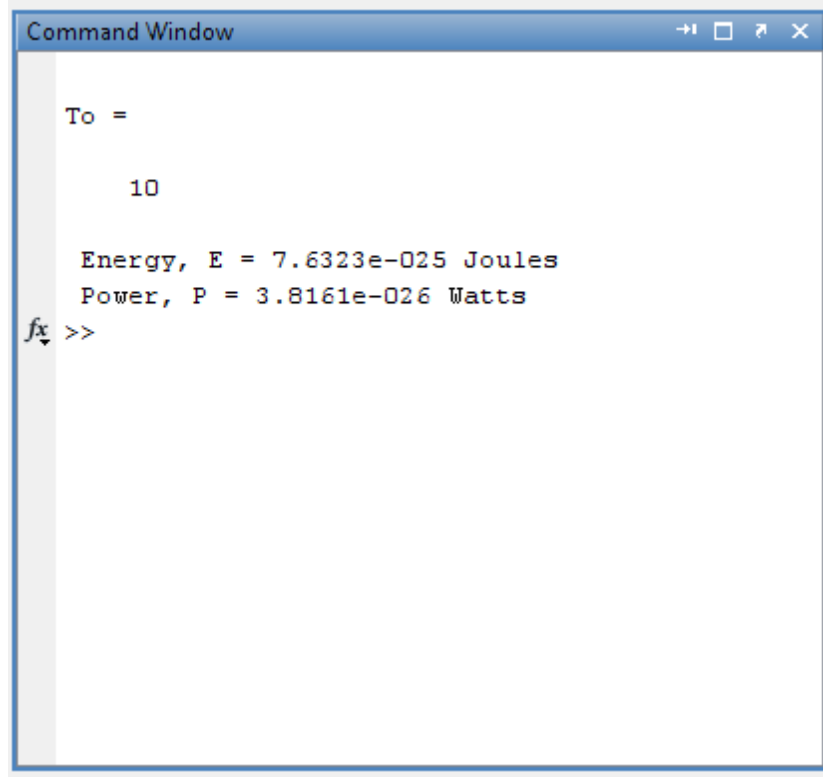
%-----computation of energy and power-----%
x=10*sin(10*pi*t);To=10
xsq=x.^2;
E = trapz(t,xsq);
P = E/(2*To);
disp([' Energy, E = ', num2str(E), ' Joules']);
disp([' Power, P = ', num2str(P), ' Watts']);

t=-10:0.1:10;
y1=sin(t);
y2=cos(t);
subplot(5,5,10),stem(t,y1),title('sinewave');grid on
subplot(5,5,11),stem(t,y2),title('coswave');grid on
%-----addition of two signals-----%
y3=y1+y2;
subplot(5,5,12),stem(t,y3),title('addition');grid on
%-----subtraction of two signals-----%
y4=y1-y2;
subplot(5,5,13),stem(t,y4),title('subtraction');grid on
%-----multiplication of two signals-----%
y5=y1.*y2;
subplot(5,5,14),stem(t,y5),title('multiplication');grid on
%-----time shifting-----%
y6=cos(t-5);
subplot(5,5,15),stem(t,y6),title('timeshifting(right)');grid on
y7=cos(t+5);
subplot(5,5,16),stem(t,y7),title('timeshifting(left)');grid on
%-----time reversal (folding) -----%
y8=sin(-t);
subplot(5,5,17),stem(t,y8),title('timefolding');grid on

%-----scaling) -----%
y9=10*sin(-t);
subplot(5,5,18),stem(t,y9),title('scaling');grid on
```

## OUTPUT:



A screenshot of the MATLAB Command Window. The window has a blue title bar with the text "Command Window" and standard window control buttons (minimize, maximize, close). The main area is white and contains the following text: "To =", "10", "Energy, E = 7.6323e-025 Joules", "Power, P = 3.8161e-026 Watts", and a prompt "fx >>".

```
Command Window

To =

    10

Energy, E = 7.6323e-025 Joules
Power, P = 3.8161e-026 Watts
fx >>
```

**RESULT:** The operations on signals and sequences: Addition, multiplication, scaling, shifting, folding, computation of energy and avg power using MATLAB



**FIND THE TRIGONOMETRIC & EXPONENTIAL FOURIER SERIES  
COEFFICIENTS OF A RECTANGULAR PERIODIC SIGNAL & PLOT ITS  
AMPLITUDE AND PHASE SPECTRUM**

**AIM:** To find the Trigonometric & exponential Fourier series coefficients

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

**PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

**PROGRAM:**

```
clear;                % clear all variables
clf;                  % clear all figures

N = 11;               % summation limit (use N odd)
wo = pi;              % fundamental frequency (rad/s)
c0 = 0;               % dc bias
t = -3:0.01:3;        % declare time values

figure(1)             % put first two plots on figure 1

% Compute yce, the Fourier Series in complex exponential form

yce = c0*ones(size(t)); % initialize yce to c0

for n = -N:2:N,       % loop over series index n (odd)
    cn = 2/(j*n*wo);   % Fourier Series Coefficient
    yce = yce + real(cn*exp(j*n*wo*t)); % Fourier Series computation
end

subplot(2,1,1)
plot([-3 -2 -2 -1 -1  0 0 1  1  2 2 3],... % plot original y(t)
```

```

        [-1 -1  1  1 -1 -1 1 1 -1 -1 1 1], ':');
hold;
plot(t,yce);                % plot truncated exponential FS
xlabel('t (seconds)'); ylabel('y(t)');
title = [' Truncated Exponential Fourier Series with N = ',...
        num2str(N)];
title(title);
hold;

% Compute yt, the Fourier Series in trigonometric form

yt = c0*ones(size(t));      % initialize yt to c0

for n = 1:2:N,              % loop over series index n (odd)
    cn = 2/(j*n*wo);        % Fourier Series Coefficient
    yt = yt + 2*abs(cn)*cos(n*wo*t+angle(cn)); % Fourier Series computation
end

subplot(2,1,2)
plot([-3 -2 -2 -1 -1  0 0 1  1  2 2 3],... % plot original y(t)
     [-1 -1  1  1 -1 -1 1 1 -1 -1 1 1], ':');
hold;                          % plot truncated trigonometric FS
plot(t,yt);
xlabel('t (seconds)'); ylabel('y(t)');
title = [' Truncated Trigonometric Fourier Series with N = ',...
        num2str(N)];
title(title);
hold;

% Draw the amplitude spectrum from exponential Fourier Series

figure(2)                      % put next plots on figure 2

subplot(2,1,1)
stem(0,c0);                   % plot c0 at nwo = 0

hold;
for n = -N:2:N,               % loop over series index n
    cn = 2/(j*n*wo);          % Fourier Series Coefficient
    stem(n*wo,abs(cn))        % plot |cn| vs nwo
end
for n = -N+1:2:N-1,          % loop over even series index n
    cn = 0;                   % Fourier Series Coefficient
    stem(n*wo,abs(cn));        % plot |cn| vs nwo
end

xlabel('w (rad/s)')
ylabel('|cn|')
title = [' Amplitude Spectrum with N = ',num2str(N)];
title(title);
grid;
hold;

```

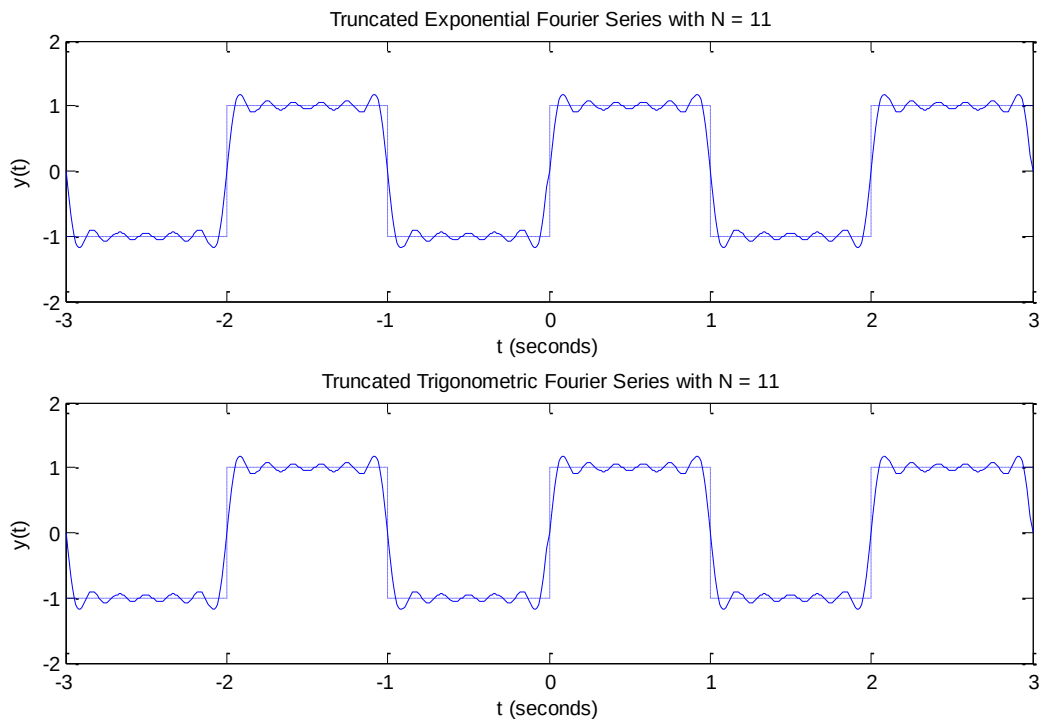
```
% Draw the phase spectrum from exponential Fourier Series

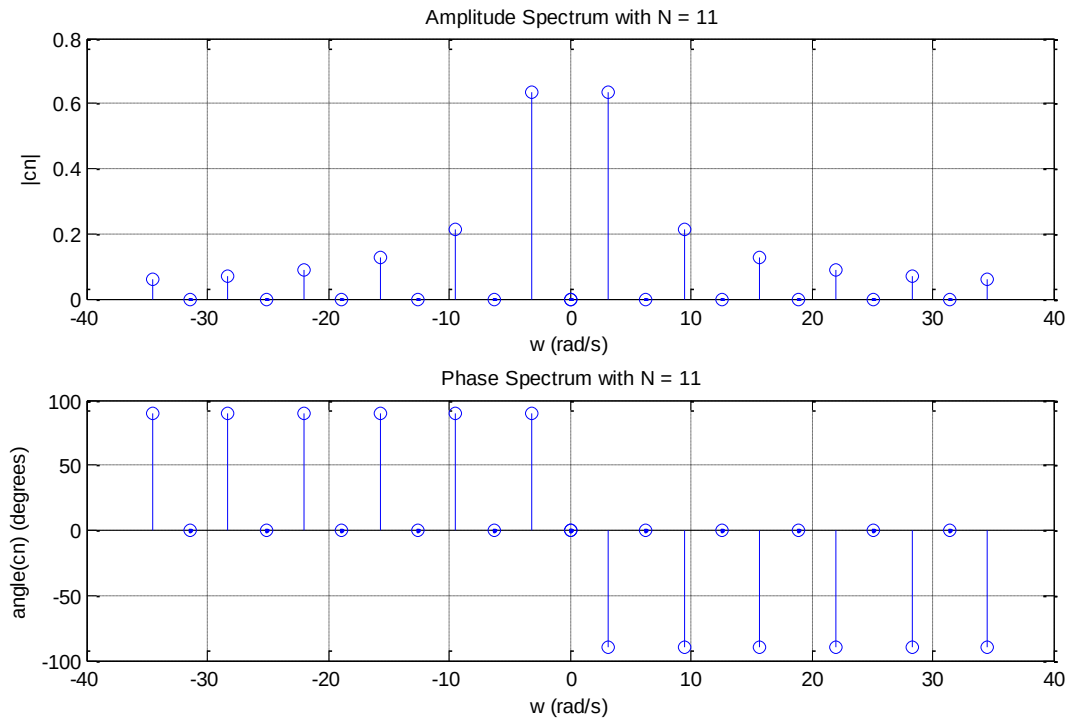
subplot(2,1,2)
stem(0,angle(c0)*180/pi);           % plot angle of c0 at nwo = 0

hold;
for n = -N:2:N,                    % loop over odd series index n
    cn = 2/(j*n*wo);               % Fourier Series Coefficient
    stem(n*wo,angle(cn)*180/pi);   % plot |cn| vs nwo
end
for n = -N+1:2:N-1,               % loop over even series index n
    cn = 0;                        % Fourier Series Coefficient
    stem(n*wo,angle(cn)*180/pi);   % plot |cn| vs nwo
end

xlabel('w (rad/s)')
ylabel('angle(cn) (degrees)')
title(['Phase Spectrum with N = ',num2str(N)]);
title(title);
grid;
hold;
```

### **OUTPUT:**





---

**RESULT:** The Trigonometric & exponential Fourier series coefficients is done using MATLAB

**FIND THE FOURIER TRANSFORM OF A GIVEN SIGNAL, PLOT ITS MAGNITUDE  
& PHASE**

**AIM:** To find the Fourier Transform of a given signal, plot its Magnitude & Phase

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

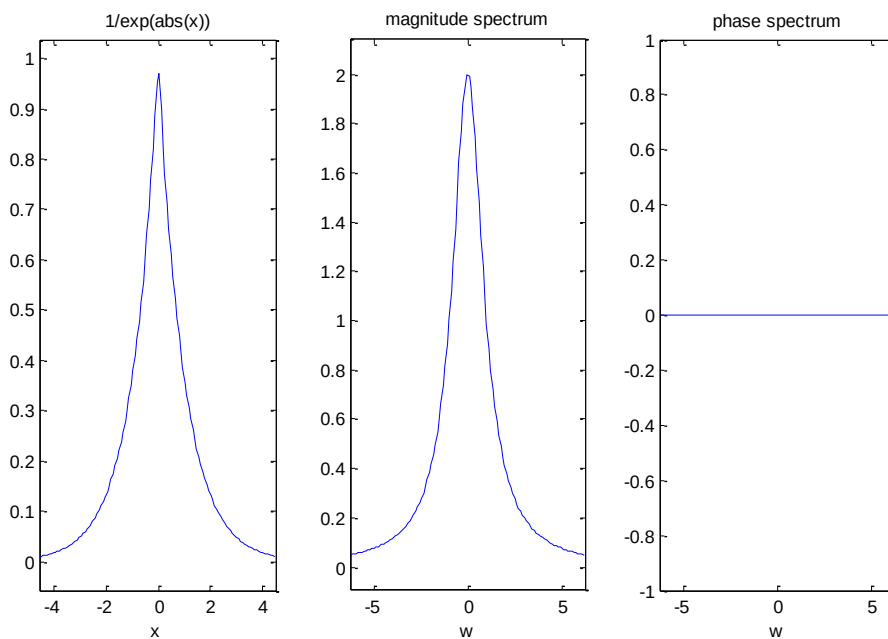
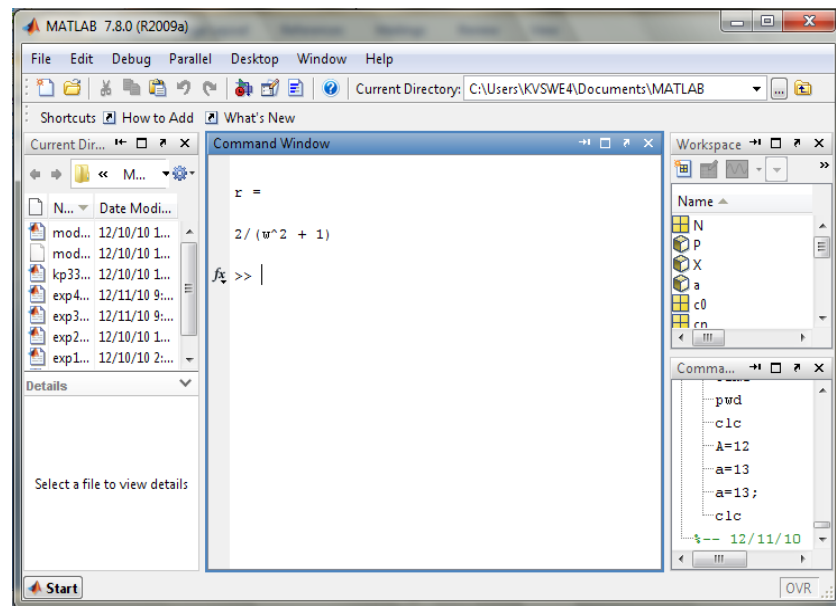
**PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

**PROGRAM:**

```
%-----Given function f(t)=e-a|x| -----%
clc;
a = sym(1);
t=0:.0002:100;
syms x
syms t
f = exp(-a*abs(x));
figure (1), subplot(1,3,1), ezplot((f))
r= (fourier(f))
xr=real(r);
xi=imag(r);
X=sqrt(xr^2+xi^2);
subplot(1,3,2), ezplot(X), title('magnitude spectrum');
P=(-atan(xi/xr));
subplot(1,3,3), ezplot(P), title('phase spectrum');
```

## OUTPUT:



**RESULT:** The Fourier Transform of a given signal, plotting its Magnitude & Phase is done using MATLAB.

**CONVOLVE TWO DISCRETE TIME SEQUENCES & PLOT**

**AIM:** To convolve two discrete time sequences & Plot.

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

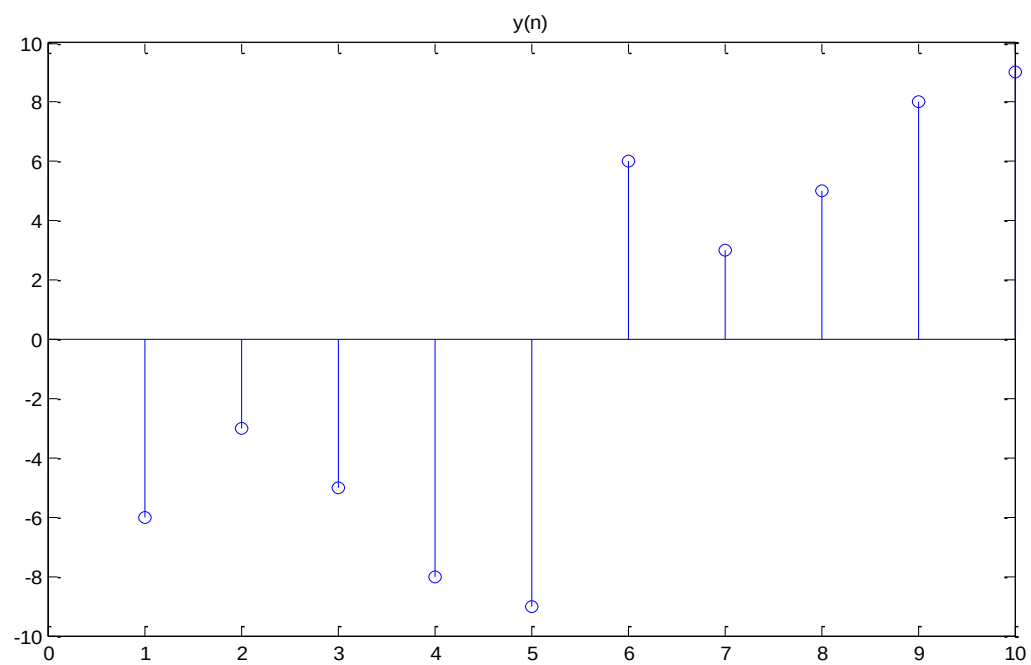
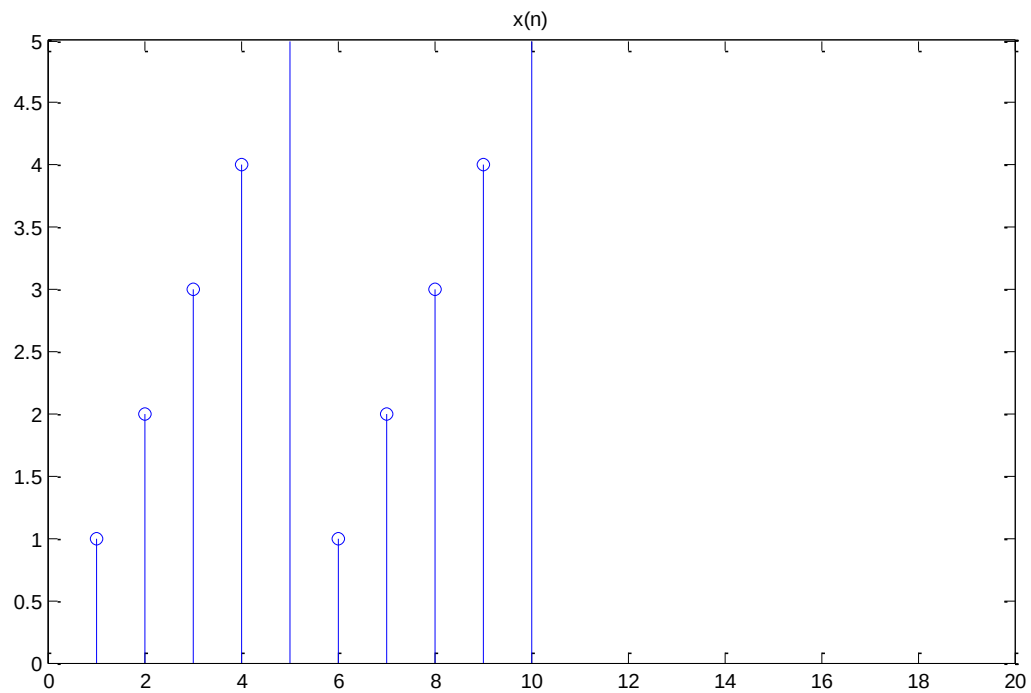
**PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

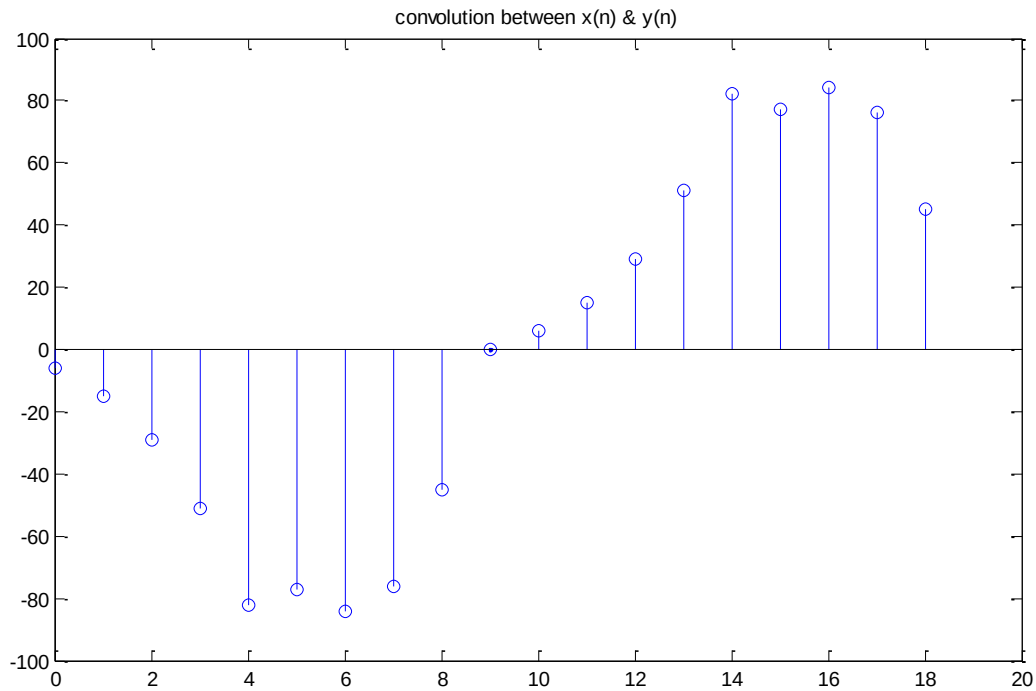
**PROGRAM:**

```
x=[1 2 3 4 5 1 2 3 4 5];  
nx= length(x);  
figure,stem(x),xlim ([0 20]);title('x(n) ');  
y=[-6 -3 -5 -8 -9 6 3 5 8 9 ];  
ny=length(y);  
figure,stem(y),xlim ([0 10]);title('y(n) ');  
h=conv(x,y);  
n3=length(h);  
n1=0:1:n3-1; %set time vector for h signal  
figure,stem(n1,h),xlim ([0 20]);title('convolution between x(n) & y(n) ');
```

**OUTPUT:**







**RESULT:** The convolution of two discrete time sequences & its Plot is done using MATLAB.

## **AUTOCORRELATION & CROSS CORRELATION DISCRETE TIME SEQUENCES & PLOT**

**AIM:** To find the autocorrelation & cross correlation of given sequences.

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

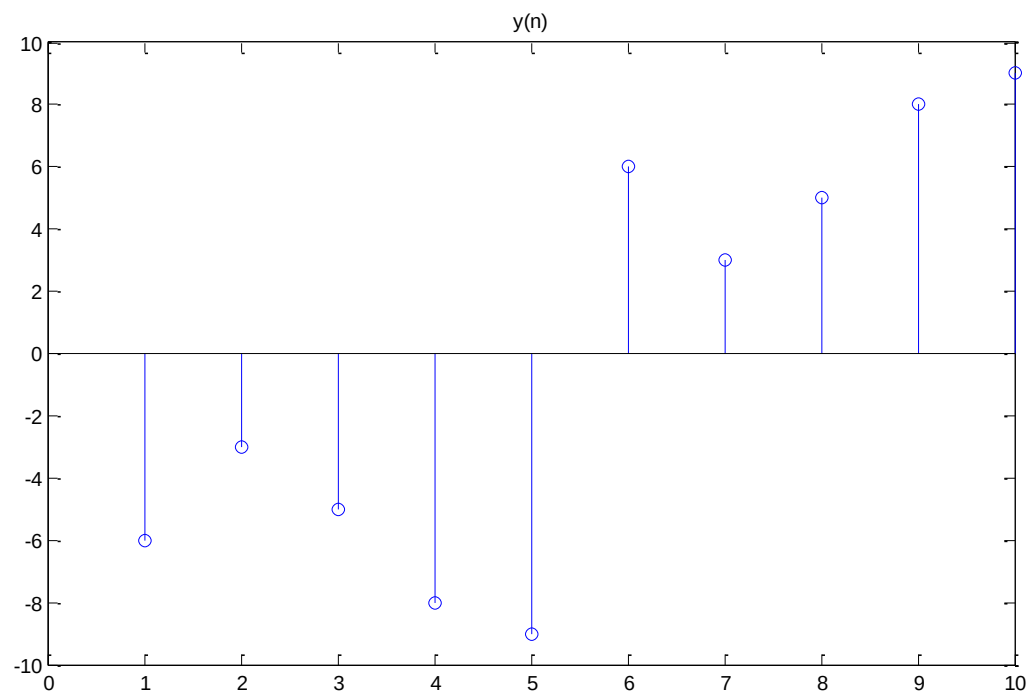
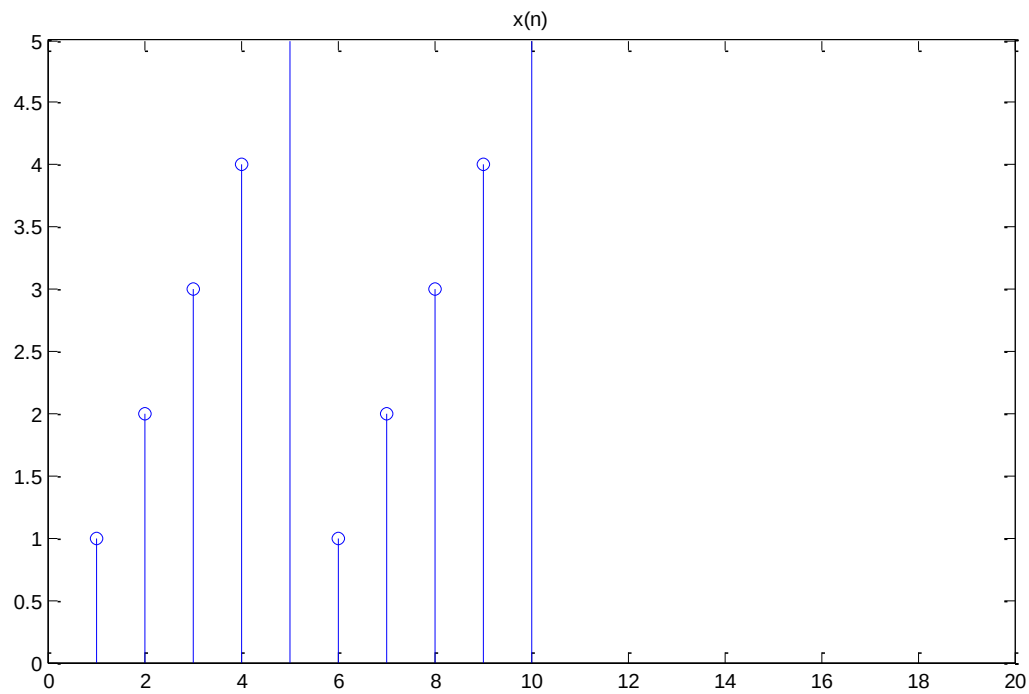
### **PROCEDURE:**

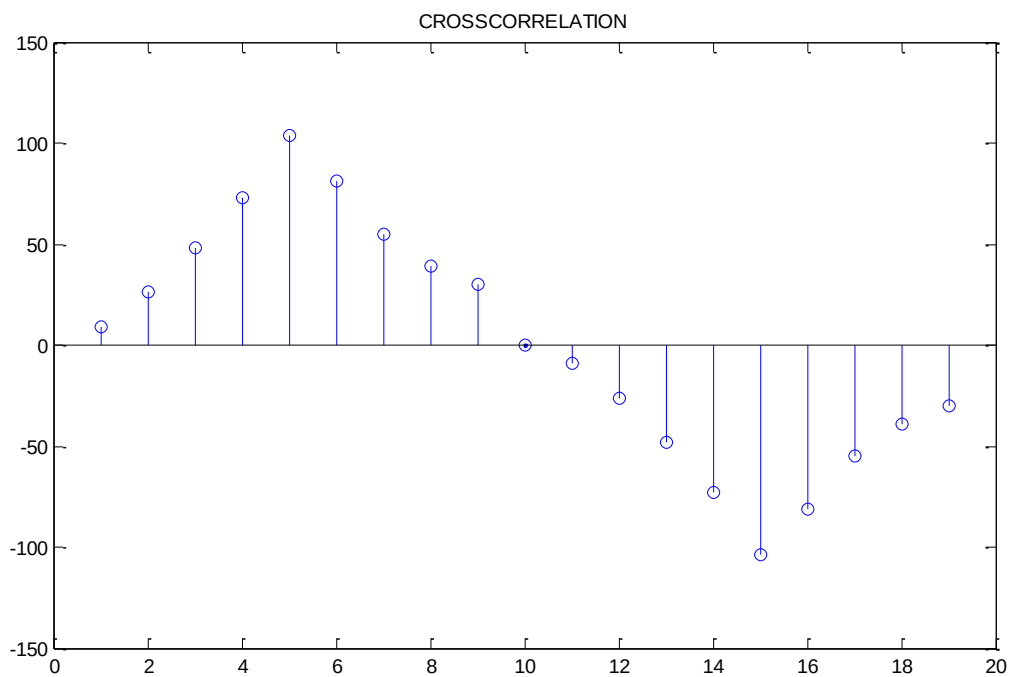
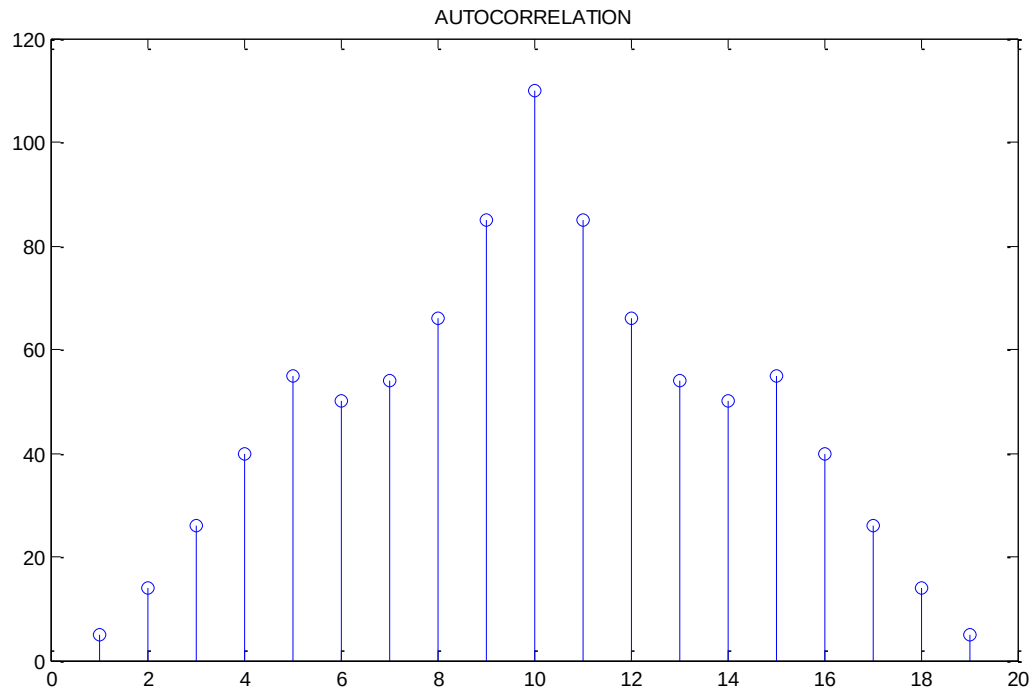
1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

### **PROGRAM:**

```
x=[1 2 3 4 5 1 2 3 4 5];
nx= length(x);
figure,stem(x),xlim ([0 20]);title('x(n) ');
y=[-6 -3 -5 -8 -9 6 3 5 8 9 ];
%ny= length(y);
figure,stem(y),xlim ([0 10]);title('y(n) ');
res1 = xcorr2(x);
figure,stem(res1);title('AUTOCORRELATION');
res2 = xcorr2(x,y);
figure,stem(res2);title('CROSSCORRELATION');
```

**OUTPUT:**





**RESULT:** The convolution of two discrete time sequences & its Plot is done using MATLAB.

**LINEARITY AND TIME INVARIANCE PROPERTIES OF SYSTEM**

**AIM:** To verify linearity and time invariance properties of system.

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

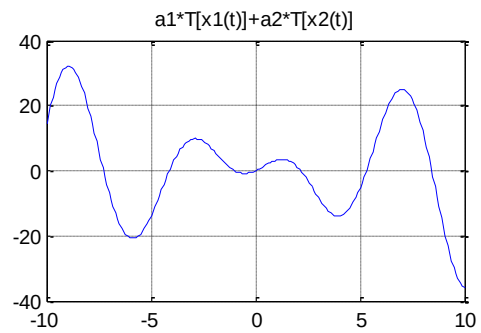
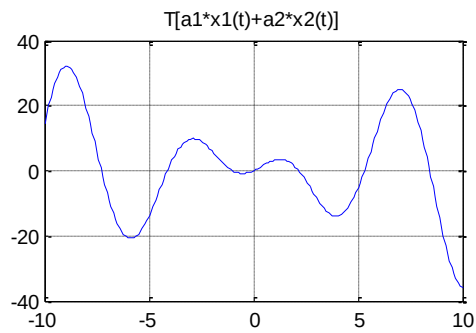
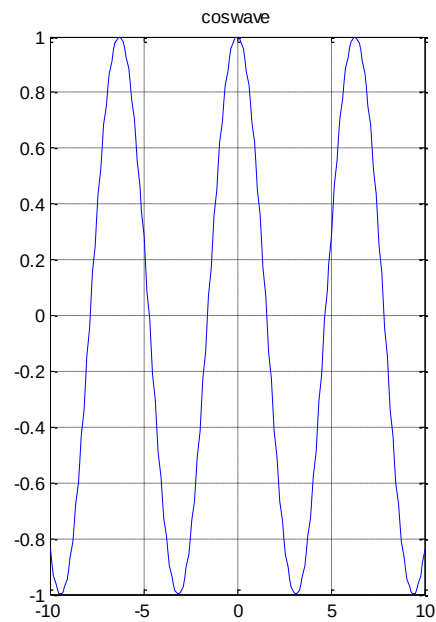
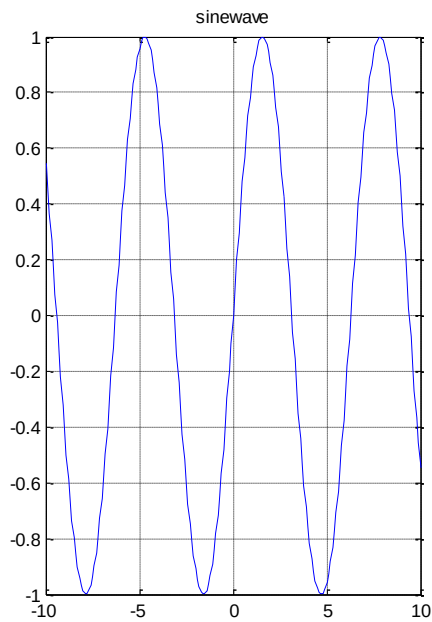
**PROCEDURE:**

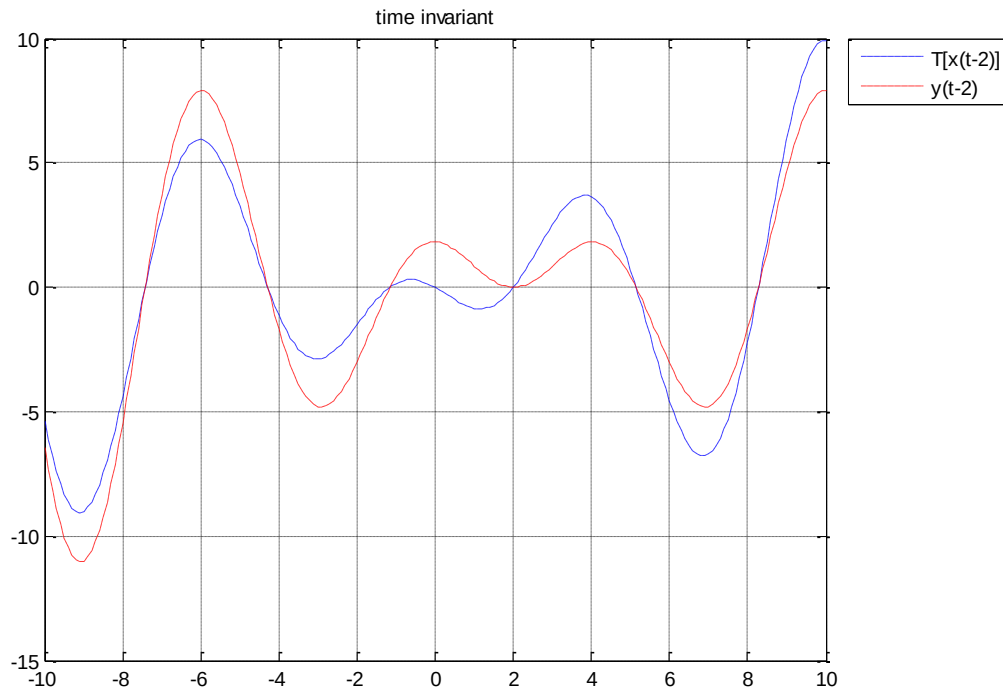
1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

**PROGRAM:**

```
t=-10:0.1:10;
y1=sin(t);
y2=cos(t);
subplot(121),plot(t,y1),title('sinewave');grid on
subplot(122),plot(t,y2),title('coswave');grid on
%-----linearity property-----%
y3=2*y1;
y4=3*y2;
y5=y3+y4;
y6=t.*y5;
y7=t.*y3;
y8=t.*y4;
y9=y7+y8;
e=(y6)-(y9);
figure,subplot(2,2,1),plot(t,y6),title('T[a1*x1(t)+a2*x2(t)]');grid on
subplot(2,2,2),plot(t,y9),title('a1*T[x1(t)]+a2*T[x2(t)]');grid on
%-----time invariant property-----%
Y2=sin(t-2);
Y3=t.*Y2;
Y4=(t-2).*sin(t-2);
figure,plot(t,Y3,'b--'),title('time invariant');grid on
, text(10*10^5,20,'level2')
hold; plot(t,Y4,'r--'),legend('T[x(t-2)]','y(t-2)',-1)
```

## OUTPUT:





**RESULT:** The linearity and time invariance properties of system are verified using MATLAB.

### **GENERATE DISCRETE TIME SEQUENCES BY SAMPLING CONTINUOUS TIME SIGNAL**

**AIM:** To generate discrete time sequences by sampling continuous time signal. Show that with sampling rates less than Nyquist rates, aliasing occurs while reconstructing the signal

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

### **PROCEDURE:**

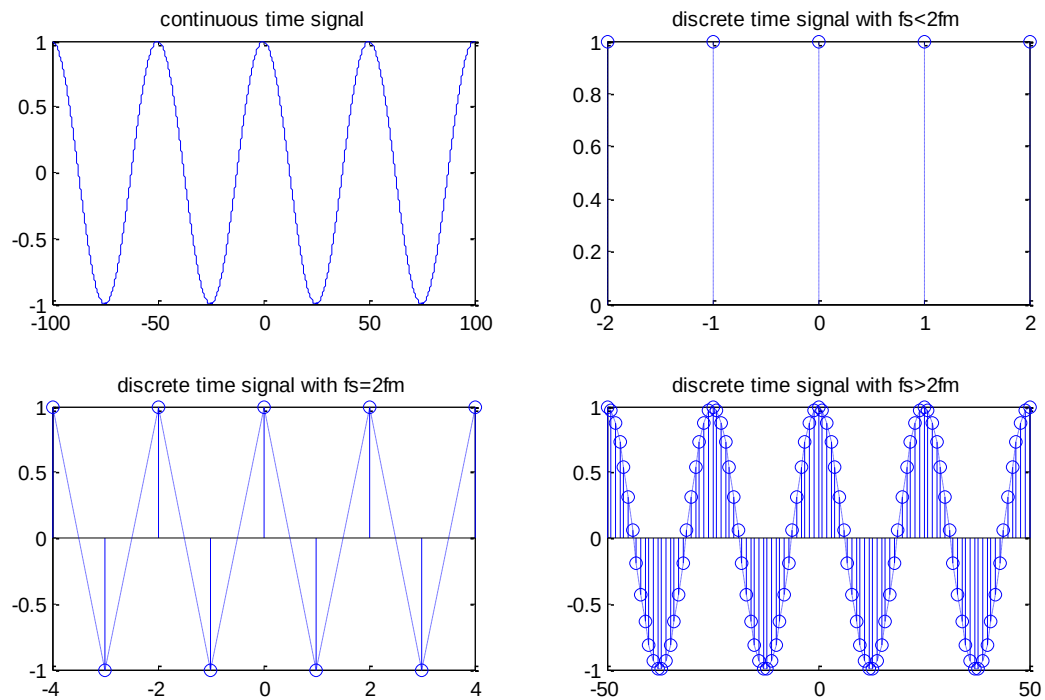
1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

### **PROGRAM:**

```
clear all;
clc;
t=-100:0.01:100;
fm=.02;
x=cos(2*pi*t*fm);
subplot(221),plot(t,x);title('continuous time signal')
fs1=.02;
n=-2:2;
x1=cos(2*pi*fm*n/fs1);
subplot(222),stem(n,x1,':'),title('discrete time signal with fs<2fm')
fs2=0.04;
n1=-4:4;
x2=cos(2*pi*fm*n1/fs2);
subplot(223),stem(n1,x2),title('discrete time signal with fs=2fm')
hold on
subplot(223),plot(n1,x2,':')
n2=-50:50;
fs3=0.5;
x3=cos(2*pi*fm*n2/fs3);
subplot(224),stem(n2,x3),title('discrete time signal with fs>2fm')
hold on
subplot(224),plot(n2,x3,':')
```



### **OUTPUT:**



**RESULT:** The generation of discrete time sequences by sampling continuous time signal is done using MATLAB.

### **FIND MAGNITUDE & PHASE RESPONSE OF FIRST ORDER LPF,HPF**

**AIM:** To find the magnitude & phase response of first order LPF,HPF. Plot the response in logarithmic scale.

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

### **PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

### **PROGRAM:**

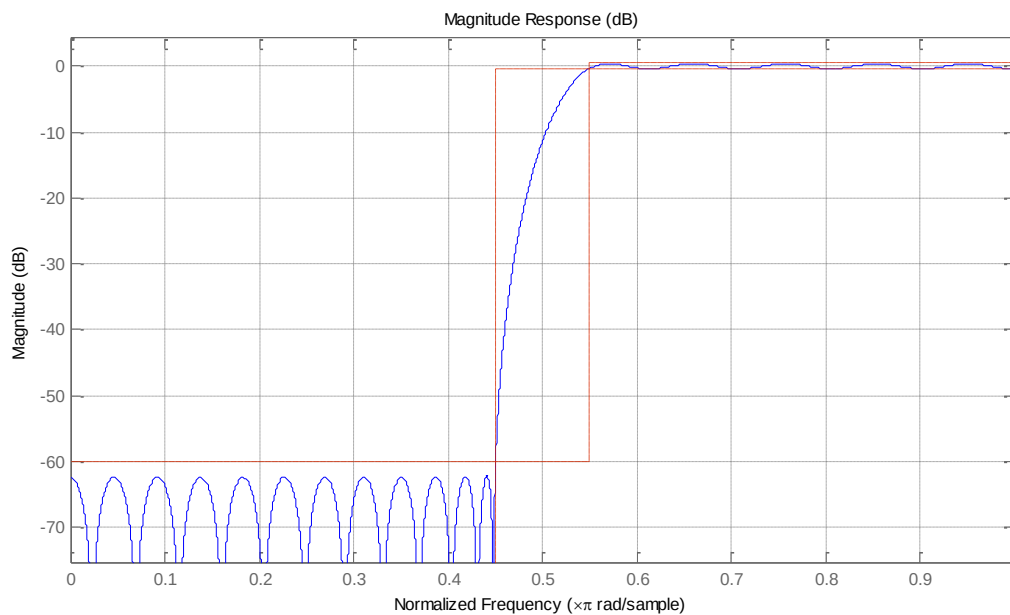
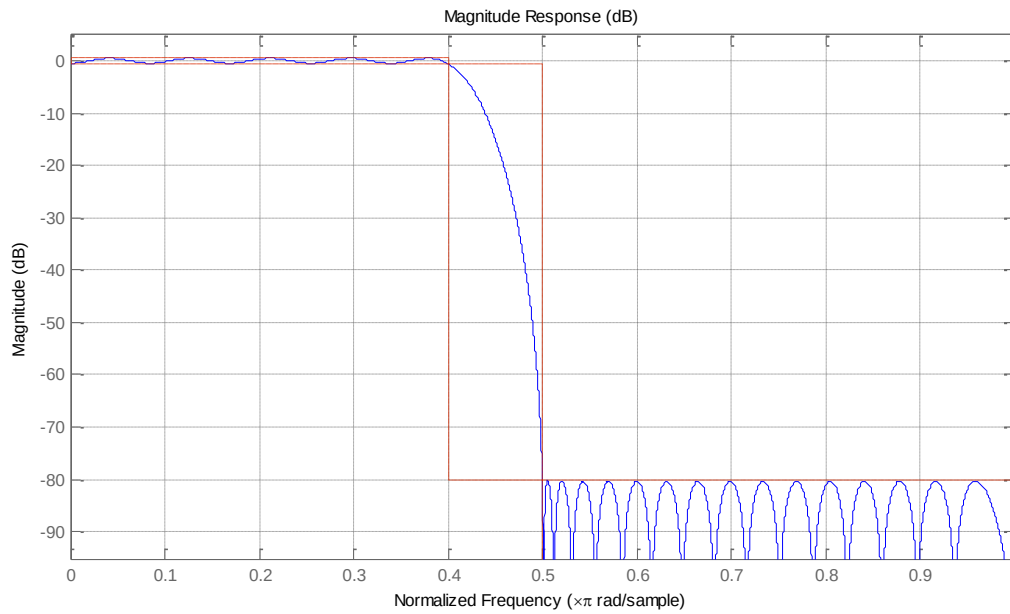
```
d = fdesign.lowpass(.4,.5,1,80)
%NormalizedFrequency: true
                %Fpass: 0.4
                %Fstop: 0.5
                %Apass: 1
                %Astop: 80

design(d)

d1 = fdesign.highpass(.45,.55,60,1)
%NormalizedFrequency: true
                %Fpass: 0.4
                %Fstop: 0.5
                %Apass: 1
                %Astop: 80

design(d1)
```

## **OUTPUT:**



**RESULT:** The magnitude & phase response of first order LPF,HPF, Plotting the response in logarithmic scale is done using MATLAB.

**FIND RESPONSE OF FIRST ORDER LPF,HPF WHEN A SPEECH SIGNAL IS  
PASSED**

**AIM:** To find the response of first order LPF,HPF when a speech signal is passed.

.

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

**PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

**PROGRAM:**

```
load chirp;
y1 = y; Fs1 = Fs;
load gong;
wavplay(y1,Fs1,'sync') % The chirp signal finishes before the
wavplay(y,Fs)           % gong signal begins playing.
L = size(y1,1);
T = 1/Fs1;
t = (0:L-1)*T;
% Input signal chart S(t)
subplot(2,3,1);
plot(t,y1);grid;
title('Signal S(t)')
xlabel('Time,s')
ylabel('Signal amplitude S(t)')
NFFT=2^16;
yf = fft(y1,NFFT)/NFFT;
f =Fs1/2*linspace(0,1,NFFT/2);
Z=2*abs(yf(1:NFFT/2));
subplot(2,3,2);
plot(f(1:NFFT/2),20*log10(Z(1:NFFT/2)));grid;
title('Signal spectrum S(t)')
```

```

xlabel('Frequency (Hz)')
ylabel('Signal magnitude |Sf(f)|, dB')
[b,a]=ellip(4,0.001,30,[300 3400]*2/Fs1);
[H,w]=freqz(b,a,NFFT);
subplot(2,3,3);
plot(w(1:NFFT)*Fs1/(2*pi),abs(H(1:NFFT)));grid;
Frequency response of the filter
title('Frequency response')
xlabel('Frequency (Hz)')
ylabel('Response factor')
SL=filter(b,a,y1);
% Output signal chart SL(t)
subplot(2,3,4);
plot(t,SL);grid;
title('Signal SL(t)')
xlabel('Time,s')
ylabel('Signal amplitude SL(t)')
% Fourier transform of the output signal (after a filtering)
SLf = fft(SL,NFFT)/NFFT;
% ff =Fs/2*linspace(0,1,NFFT/2);
ZZ=2*abs(SLf(1:NFFT/2));
% Output (filtered) signal spectrum chart SLf(f)
subplot(2,3,5);
plot(f(1:NFFT/2),20*log10(ZZ(1:NFFT/2)));grid;
title('Signal spectrum SL(t)')
xlabel('Frequency (Hz)')
ylabel('Signal magnitude |SLf(f)|, dB')
% Write to disk
wavwrite(SL,Fs,'output.wav');
Powf1 = trapz(f,Z.^2)
Powf2 = trapz(f,ZZ.^2)
Sp1 = trapz(t,y1.^2)
Sp2 = trapz(t,SL.^2)
Ratio = Powf2/Powf1

d = fdesign.lowpass(.4,.5,1,80)
%NormalizedFrequency: true
           %Fpass: 0.4
           %Fstop: 0.5
           %Apass: 1
           %Astop: 80

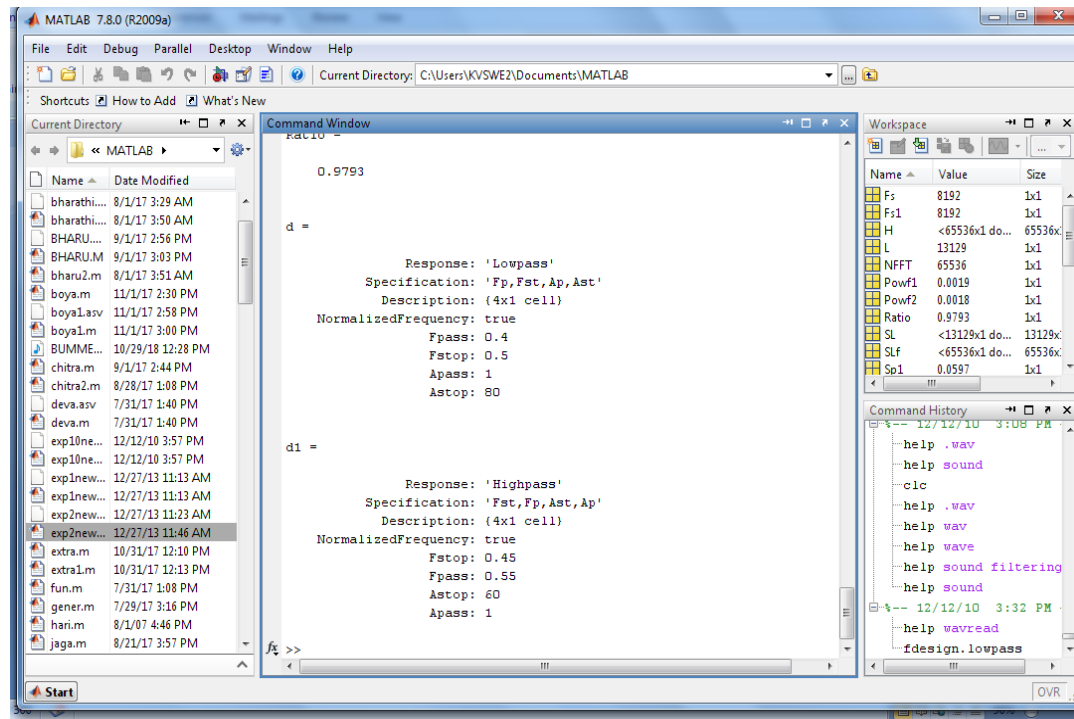
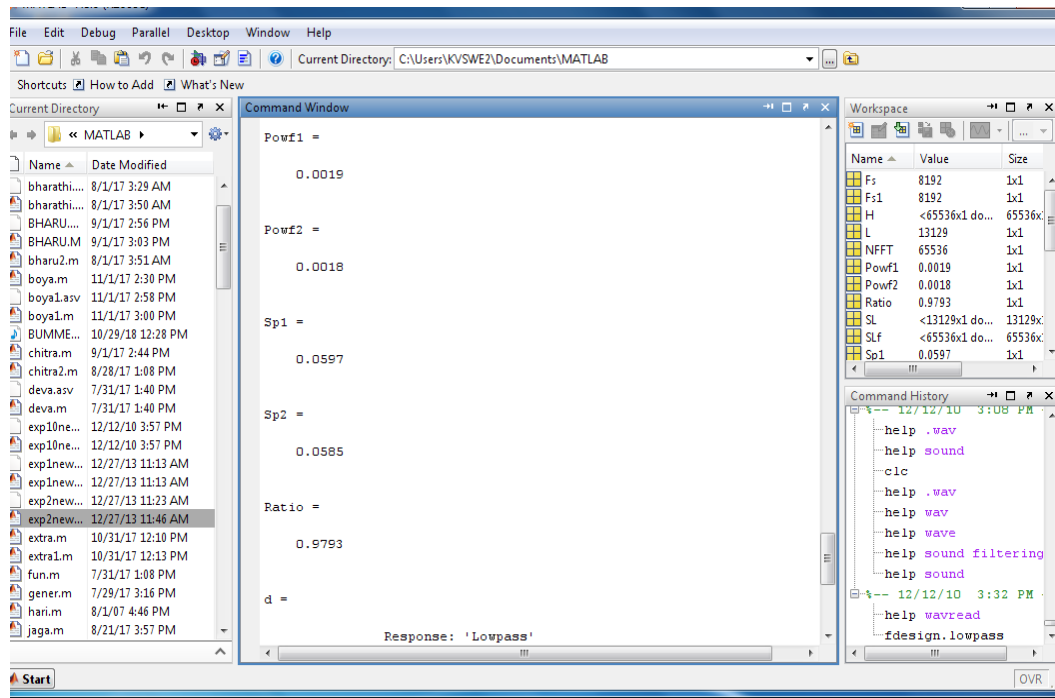
design(d)

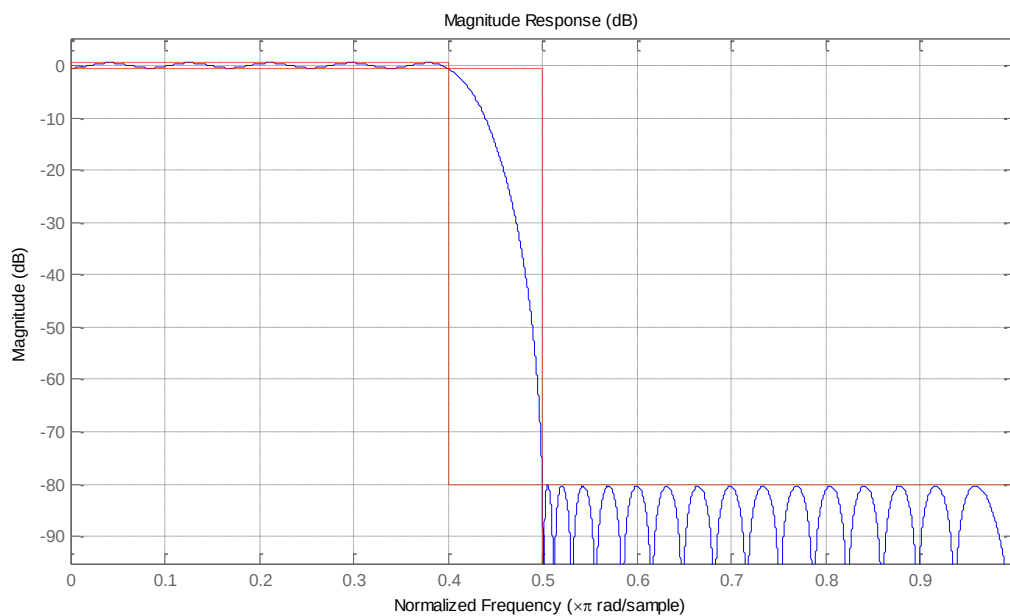
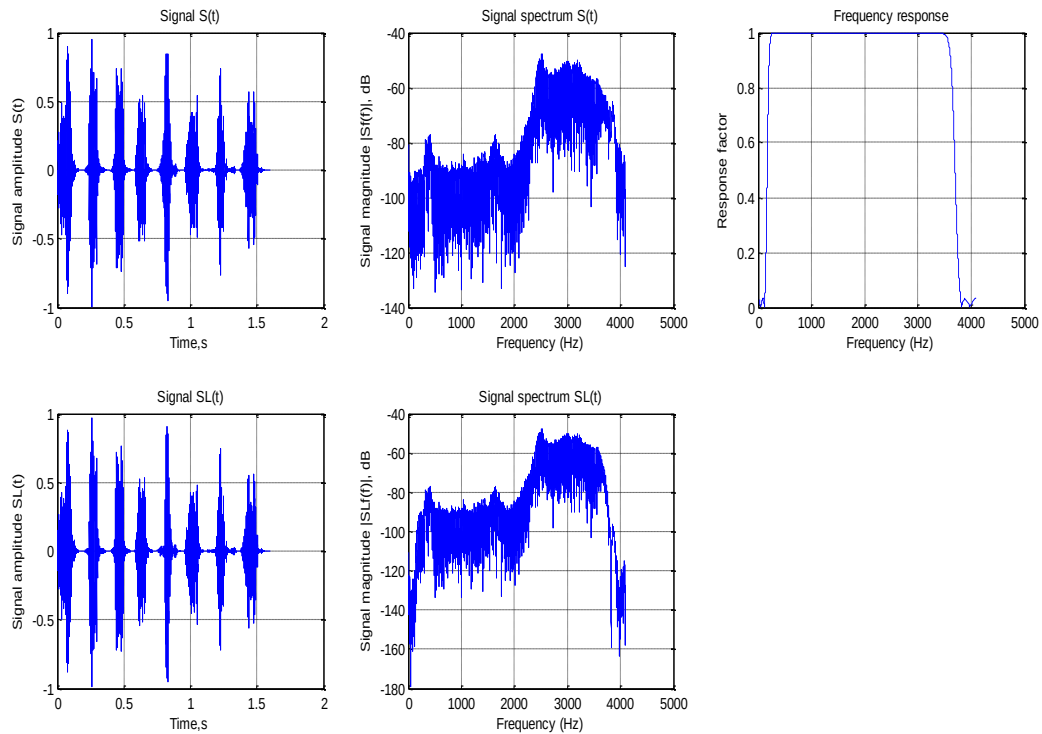
d1 = fdesign.highpass(.45,.55,60,1)
%NormalizedFrequency: true
           %Fpass: 0.4
           %Fstop: 0.5
           %Apass: 1
           %Astop: 80

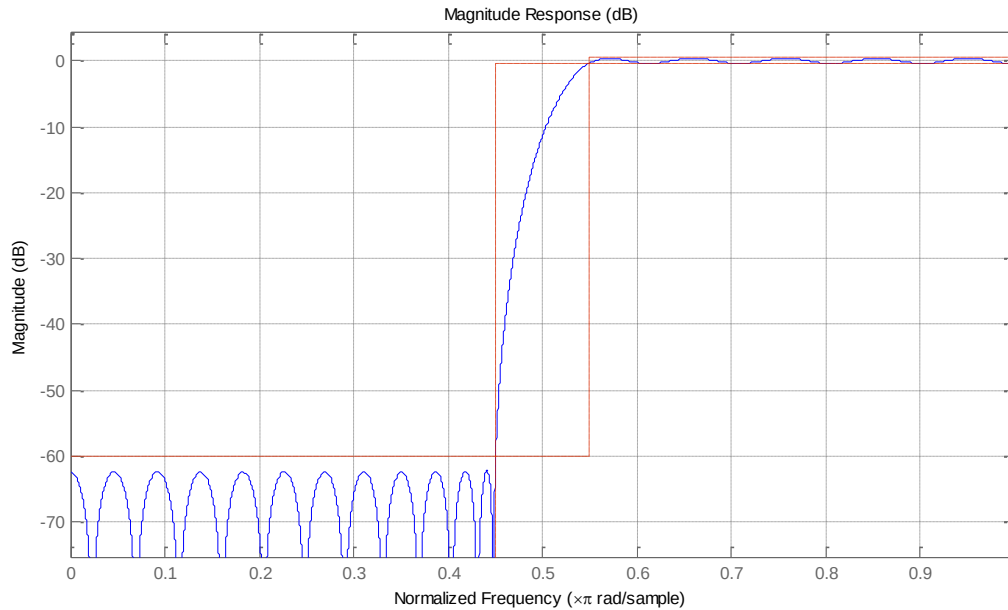
design(d1)

```

## OUTPUT:







**RESULT:** The magnitude & phase response of first order LPF,HPF, Plotting the response in logarithmic scale is done using MATLAB.



**COMPLEX GAUSSIAN NOISE ITS MEAN VARIANCE , PROBABILITY DENSITY  
FUNCTION & POWER SPECTRAL DENSITY**

**AIM:** To Generate Complex Gaussian Noise Find its Mean Variance, Probability Density Function & Power Spectral Density.

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

**PROCEDURE:**

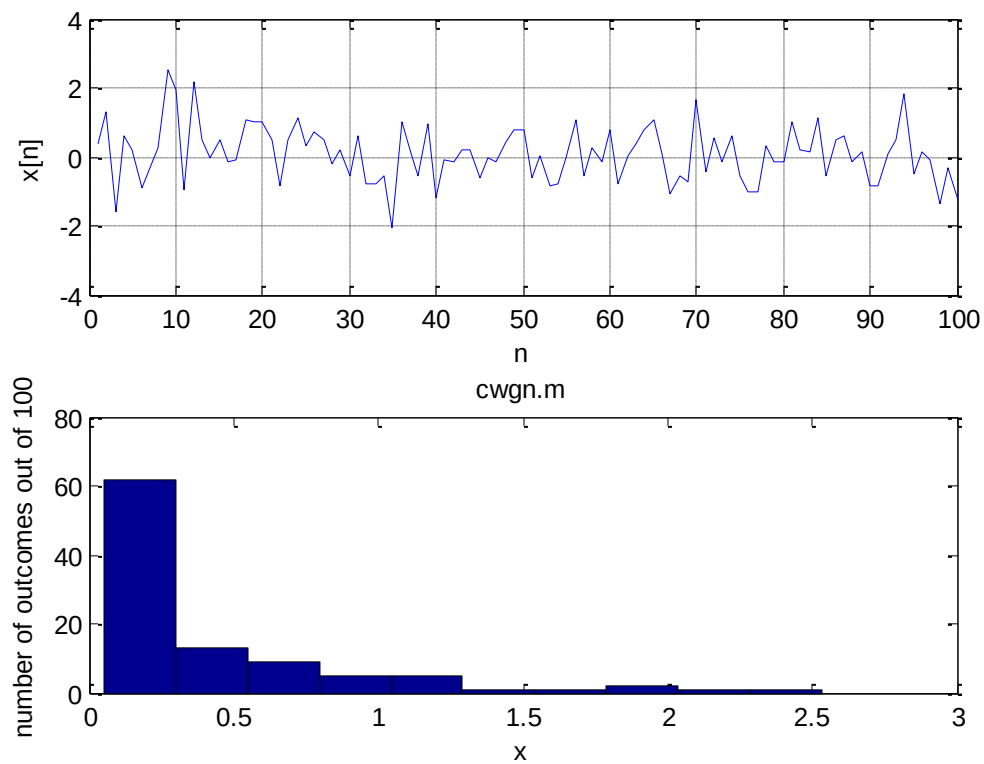
1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

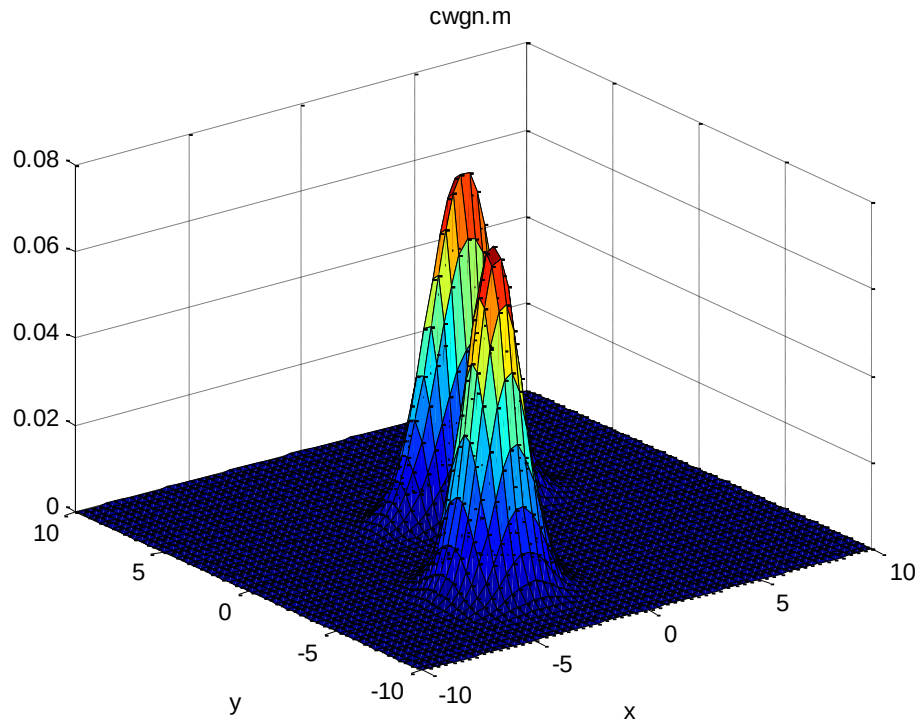
**PROGRAM:**

```
% cwgn.m
%
% This program generates complex white Gaussian noise and
% then estimates its mean and variance.
% N=100;
N=100;
t=1:1:100
varw=1;
x=sqrt(varw/2)*randn(N,1)+j*sqrt(varw/2)*randn(N,1);
muest=mean(x)
varest=cov(x)
% estimated PDF for real white Gaussian noise. randn('state',0) %
x=randn(100,1);
subplot(2,1,1), plot(t,x);
xlabel('n'); ylabel('x[n]'); grid
subplot(2,1,2)
hist(x);
xlabel('x')
ylabel('number of outcomes out of 100')
title('cwgn.m')
figure
```

```
mu = [1 2;-3 -5];  
sigma = cat(3,[2 0;0 .5],[1 0;0 1]);  
p = ones(1,2)/2;  
obj = gmdistribution(mu,sigma,p);  
  
ezsurf(@(x,y)pdf(obj,[x y]),[-10 10],[-10 10])  
  
%pdf(x,100,10,-3,3,1)  
%xlabel('x'); ylabel('PDF, p(x)');  
title('cwgn.m')
```

### **OUTPUT:**





```
Command Window

Columns 92 through 98

    92    93    94    95    96    97    98

Columns 99 through 100

    99   100|

muest =

    0.0870 - 0.0514i

varest =

    1.1806
```

**RESULT:** Complex Gaussian Noise is generated and its Mean Variance, Probability Density Function & Power Spectral Density is done using MATLAB.

### **GENERATE RANDOM BIPOLAR DATA WITH DATA RATE AND PLOT FOR TIME PERIOD**

**AIM:** To generate random bipolar data with data rate 10kbps and plot for time period of 0.2 sec.

**APPARATUS:** Computer

MATLAB 7.8.0.347(R2009a) Software.

### **PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

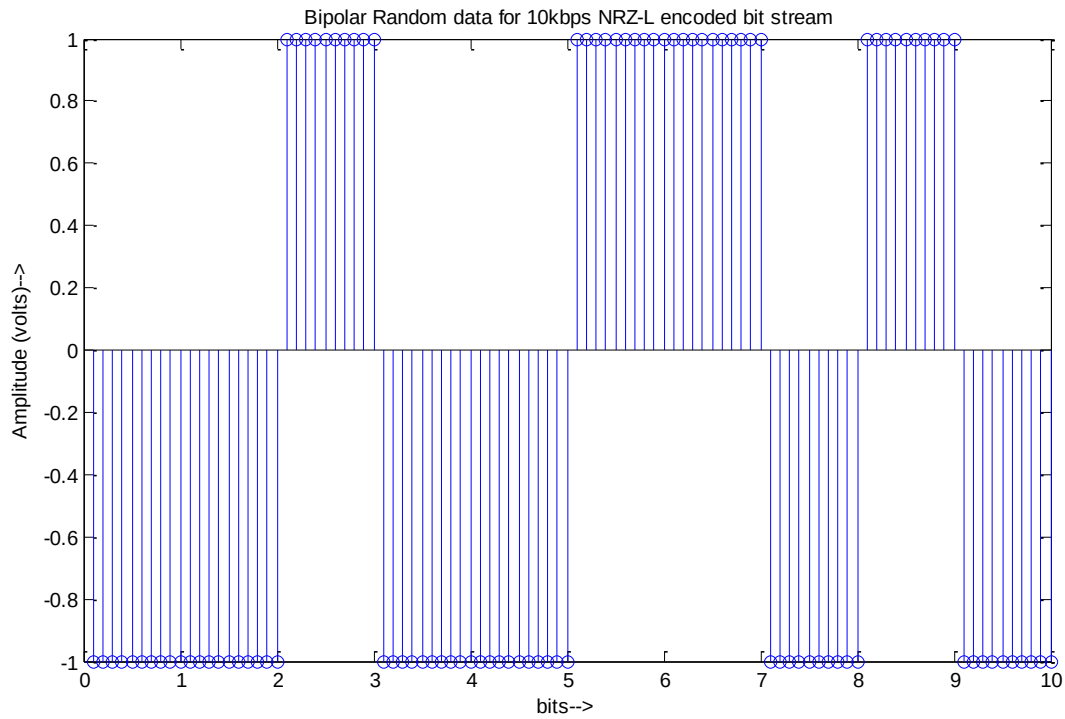
### **PROGRAM:**

```
clear all;
close all;
%Nb is the number of bits to be transmitted
Nb=10;
% Generate Nb bits randomly
b=rand(1,Nb)>0.5;
%Rb is the bit rate in bits/second
Rb=10000;

fs=50;    % For T=0.2s fs =50Hz%
NRZ_out=[];

%Vp is the peak voltage +v of the waveform
Vp=1;
%Here we encode input bitstream as Bipolar NRZ-L waveform
for index=1:size(b,2)
    if b(index)==1
        NRZ_out=[NRZ_out [1 1 1 1 1 1 1 1 1 1]*Vp];
    elseif b(index)==0
        NRZ_out=[NRZ_out [1 1 1 1 1 1 1 1 1 1]*(-Vp)];
    end
end
```

```
figure;  
stem((1:Nb*10)/10,NRZ_out);  
xlabel('bits-->');  
ylabel('Amplitude (volts)-->');  
title(' Bipolar Random data for 10kbps NRZ-L encoded bit stream');
```



**OUTPUT :** Random Bipolar data is Generated with data rate 10kbps and plotted for time period of 0.2 sec is done in MATLAB.

### **TO PLOT POLE-ZERO DIAGRAM IN S-PLANE /Z PLANE OF GIVEN SIGNAL/SEQUENCE AND VERIFY ITS STABILITY**

**AIM:** To plot pole-zero diagram in S-Plane /Z plane of given Signal/Sequence and verify its Stability

**APPARATUS:**        Computer  
                         MATLAB 7.8.0.347(R2009a) Software.

### **PROCEDURE:**

1. Open the MATLAB software by double clicking the ICON on desktop.
2. Open the new M-file by using file menu
3. Write the program in new file
4. Click on save and run the icon
5. Perform error check which displayed on command window
6. Plot the waveforms which displays on figure window.
7. Note down the values, which displays on the work space.

### **PROGRAM:**

```
clear;                                % clear matlab memory

numH = [2 5 7];                      % numerator of H(s)
denH = [3 1 4 9];                   % denominator of H(s)

disp(' '); disp('H(s) = '); % print H(s) to window
printsys(numH,denH,'s');

% Plot system poles and zeros
figure(1); clf;                      % put plots in figure 1
pzmap(numH,denH);

% Compute and Plot step/impulse responses

figure(2); clf;                      % put plots in figure 2

t = 0:0.01:10;
y = step(numH,denH,t);               % compute & plot system step response
```

```

subplot(2,1,1);
plot(t,y);
xlabel('time <sec>');
ylabel('y(t)');
title('System Step Response');

y = impulse(numH,denH,t); % compute & plot impulse response
subplot(2,1,2);
plot(t,y);
xlabel('time <sec>');
ylabel('y(t)');
title('System Impulse Response');

% Compute and Plot Spectra of H(w) using freqs

figure(3); clf; % put plots in figure 3

w = -20:0.05:20;
H = freqs(numH,denH,w); % compute and plot H(w) spectra
subplot(2,2,1);
plot(w,abs(H));
xlabel('w <rad/sec>');
ylabel('|H(w)|');
title('Magnitude Spectrum of Impulse Response');

subplot(2,2,2);
plot(w,angle(H)*180/pi);
xlabel('w <rad/sec>');
ylabel('Phase(H(w)) <deg>');
title('Phase Spectrum of Impulse Response');

% Compute and Plot Spectra of H(w) in log form using freqs

w = logspace(-1,1,100); % w has 100 values from 10^(-1) to 10^1
                        % with same # of points in each decade

H = freqs(numH,denH,w); % compute and plot H(w) spectra
subplot(2,2,3);
semilogx(w,20*log(abs(H)));
xlabel('w <rad/sec>');
ylabel('|H(w)| <dB>');
title('Magnitude Spectrum of Impulse Response');

subplot(2,2,4);
semilogx(w,angle(H)*180/pi);
xlabel('w <rad/sec>');
ylabel('Phase(H(w)) <deg>');
title('Phase Spectrum of Impulse Response');

% Compute and Plot Spectra of H(w) using bode

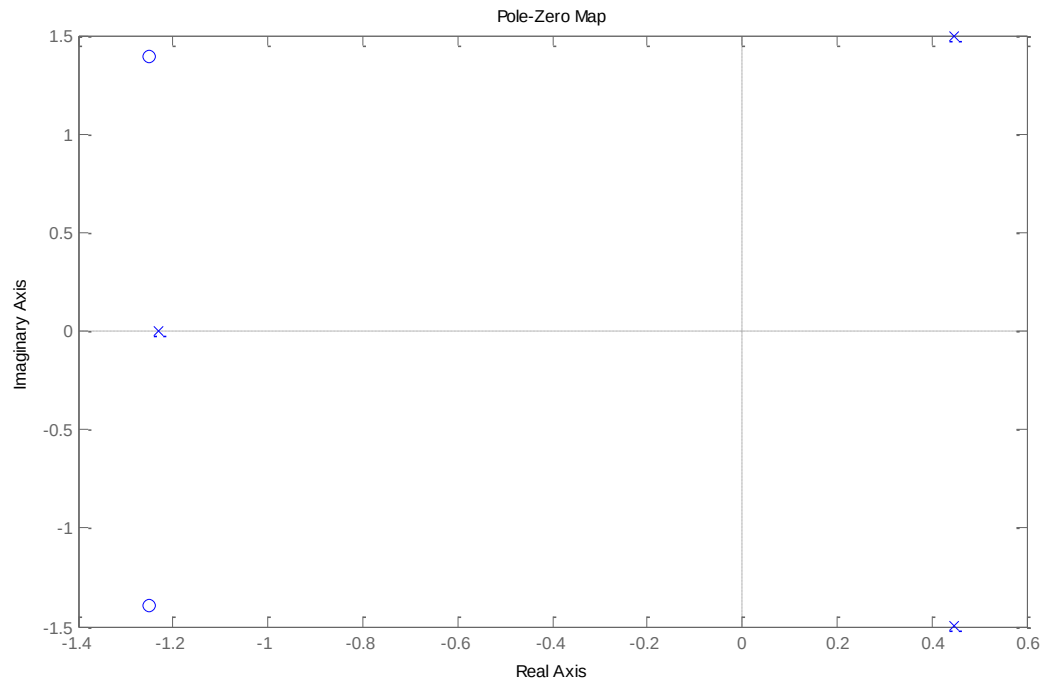
figure(4); clf; % put plots in figure 4
g = tf([1 0.1 7.5],[1 0.12 9 0 0]);

```

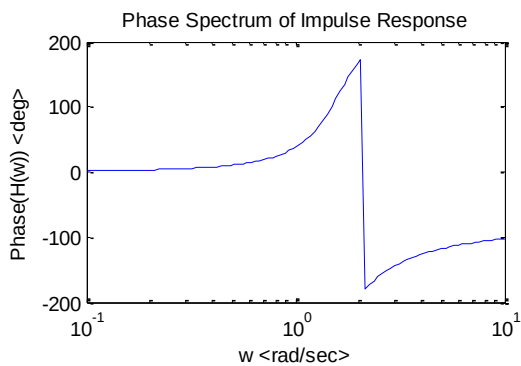
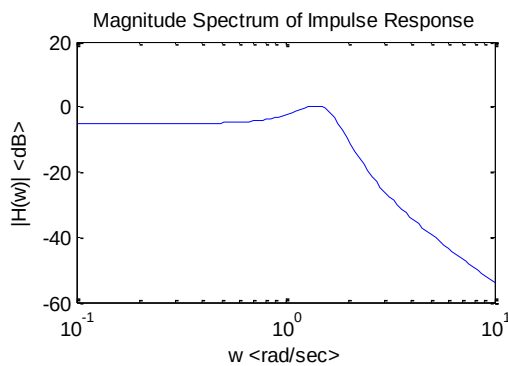
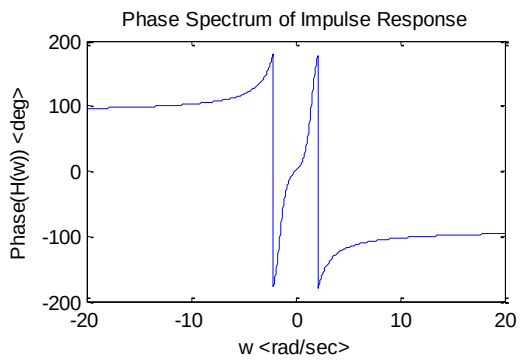
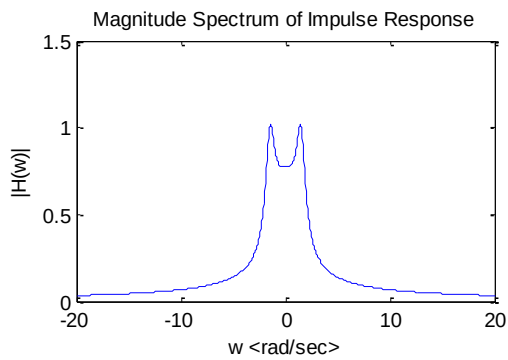
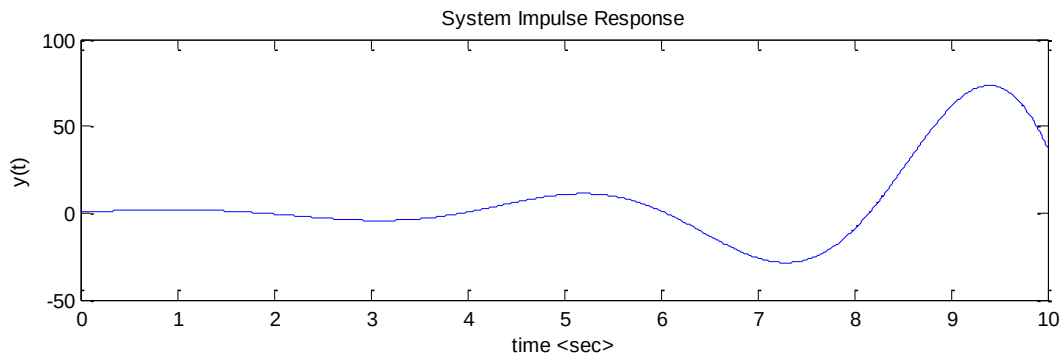
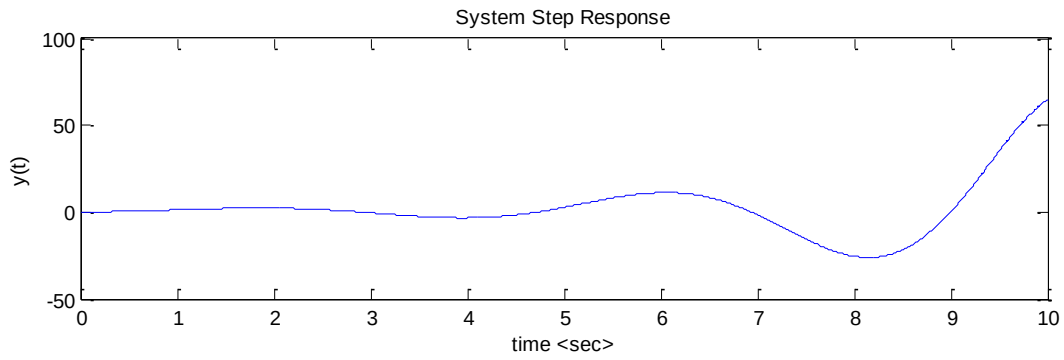
bode (g)

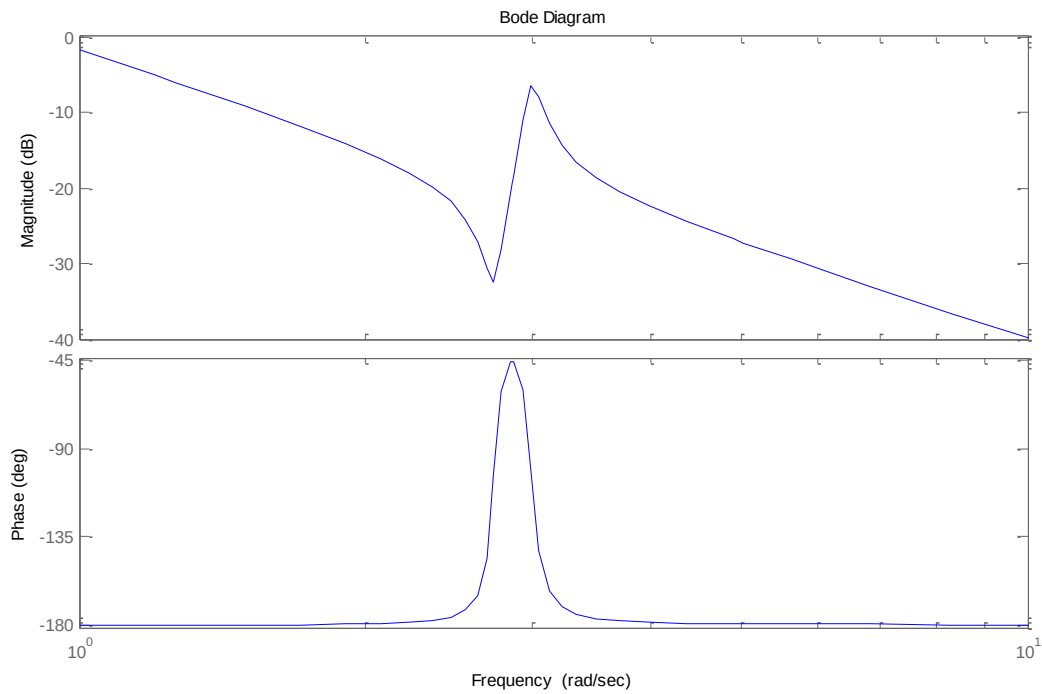
% Compute and Plot Spectra of  $H(w)$  in log form using bode

### **OUTPUT :**









**OUTPUT :** Pole-zero diagram in S-Plane /Z plane of given Signal/Sequence and verified its Stability is done in MATLAB.

# **ADD-on Experiments**

## Experiment No: 1

### Generate a simple text image using MATLAB

**AIM:** To generate a simple ECE text/binary imageby using MATLAB.

**SOFTWARE:** Personal Computer with windows 7 operating system loaded with MATLAB R2009a

**PROGRAM:**

```
clc;
clear all;
close all;

s=[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0;
0 255 255 255 255 0 255 255 255 255 0 255 255 255 255 0 0
0;
0 255 0 0 0 0 255 0 0 0 0 255 0 0 0 0 0
0;
0 255 255 255 255 0 255 0 0 0 0 255 255 255 255 0 0
0;
0 255 0 0 0 0 255 0 0 0 0 255 0 0 0 0 0
0;
0 255 255 255 255 0 255 255 255 255 0 255 255 255 255 0 0
0;
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0;]

imshow(s);
```

**OUTPUT:**



**RESULT:** Generation of a simple ECE text/binary imageby using MATLAB is done.

### Experiment No: 2

#### Insert text image in image using MATLAB

**AIM:** Insert the ECE text/binary image in another image by using MATLAB.

**SOFTWARE:** Personal Computer with windows 7 operating system loaded with MATLAB R2009a

**PROGRAM:**

```
clc;
clear all;
close all;

s=[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0;
0 255 255 255 255 0 255 255 255 255 0 255 255 255 255 0 0
0;
0 255 0 0 0 0 255 0 0 0 0 255 0 0 0 0 0
0;
0 255 255 255 255 0 255 0 0 0 0 255 255 255 255 0 0
0;
0 255 0 0 0 0 255 0 0 0 0 255 0 0 0 0 0
0;
0 255 255 255 255 0 255 255 255 255 0 255 255 255 255 0 0
0;
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0;]

imshow(s);

[a b]=size(s);
imshow(s);

I = imread('cameraman.tif');
I1=uint8(s)

I(1:a,1:b)=I1;
imshow(I);
```

**OUTPUT:** Insertion of the ECE text/binary image in another image by using MATLAB is done.



**RESULT:**